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Elucidating the behavior, kinetics and mechanism of wet carbonation in ternesite

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ABSTRACT

In this study, the carbonation kinetics, phase development, and microstructure evolution of ternesite ($C_5S_2\$$) phase were investigated via wet carbonation. The results indicated that ternesite exhibited remarkable carbonation reactivity, achieving approximately 95% completion of the carbonation reaction within 15 min and facilitating CO_2 sequestration exceeding 340 g per kilogram of ternesite. The main carbonation products identified were aragonite, calcite, gypsum, and silica gel (Si-gel). The carbonation kinetics conformed to the surface coverage model, with the reaction proceeding through distinct stages of acceleration, deceleration, and stabilization, ultimately resulting in the formation of a "core-shell" structure. Notably, the crystalline phase of calcium carbonate (CC) underwent spontaneous transformation during the carbonation of ternesite. It is observed that CC exhibited a polycrystalline structure and underwent a distinct phase transformation from calcite to aragonite. The resulting CC was predominantly metastable, exhibiting low crystallinity and reduced crystallite size. These characteristics are attributed to the intrinsic properties of ternesite, particularly the presence of SO_4^{2-} ions, which promote the formation and stabilization of metastable aragonite. Furthermore, the changes in crystalline and morphology of CC, induced by dissolution and re-precipitation process, disrupted the CC layer and thereby facilitated the release of another significant product, Si-gel. These distinctive carbonation behaviors suggest that ternesite has potential for broader applications, including its use as a supplementary cementitious material or as a precursor for the synthesis of high value-added products such as aragonite whiskers and nano-sized SiO_2.

1. Introduction

The cement industry contributes approximately 8% of anthropogenic CO_2 emissions, and its deep decarbonation remains a critical obstacle to achieving global carbon neutrality goals [1,2]. The carbon footprint of cement production has a well-defined structure with approximately 50% derived from carbonate decomposition during limestone calcination, 40% originates from fossil fuel combustion, and the remaining 10% arises from indirect emissions associated with electricity generation and transportation [3–5]. Consequently, cement decarbonization must move beyond traditional single-track strategies focused solely on energy-efficiency enhancements, advancing instead toward systemic innovations that integrate source-level carbon reduction with end-use carbon utilization [6,7]. Source-end carbon reduction aims to decrease limestone dependence and lower calcination energy demand through innovations in the clinker chemical system [8,9]. End-use carbon leverages Carbon Capture, Utilization, and Storage (CCUS) technology to convert emitted CO_2 into stable carbonates, thereby enabling potential negative-emission pathways [10–12]. Against this backdrop, clinker minerals that combine low-calcium synthesis requirements with high-efficiency carbonation reactivity have emerged as a major research focus.

The development of new low-calcium clinker represents a fundamental pathway for source-level carbon reduction. Portland cement clinker, dominated by alite (C_3S), requires formation temperatures as high as 1450 °C and relies heavily on high-purity limestone, thereby sustaining a high carbon-emission intensity [13]. In contrast, low-lime minerals directly reduce carbonate decomposition emissions by lowering their CaO content and provide additional energy savings through their lower firing temperatures [14,15]. Current research

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focuses on typical low-lime minerals including larnite (β-C_2S), calcio-olivine(γ-C_2S), rankinite (C_3S_2), wollastonite (CS), and ternesite [16–19]. The key physicochemical indicators of these minerals are listed in Table 1. Among them, ternesite ($4CaO{\cdot}2SiO_2{\cdot}CaSO_4$, abbreviated as $C_5S_2\$$), a transitional phase formed during sulfoaluminate clinker synthesis, is regarded as a clinker mineral with notable low-carbon potential [20,21]. The formation and stability of ternesite are highly dependent on temperature, it can be synthesized and stabilized via the secondary reaction of belite and gypsum within the temperature range of 1100–1200 °C [22,23]. As shown in the Table 1, ternesite exhibits a lower calcination temperature and a reduced limestone requirement, thereby offering significant carbon emission advantages [24–26]. More importantly, ternesite can be prepared from low-grade limestone, clay, and industrial by-product gypsum, which facilitates resource recycling and cost reduction [27–29]. However, similar to other low-lime minerals, ternesite exhibits inherently low hydration activity, which hinders the development of satisfactory mechanical properties under conventional hydration curing [20,22]. Previous studies have demonstrated that the hydration activity of ternesite can be enhanced through synergistic interactions with other aluminate minerals [30,31]. Additionally, tailored liquid environments and optimized curing regimes have been shown to accelerate the dissolution and hydration processes of ternesite [32,33]. Despite these advances, the practical application and allowable dosage of ternesite remain substantially constrained by its limited reactivity and the stringent environmental conditions required for its effective use.

Meanwhile, ternesite exhibits remarkable CO_2 mineralization (i.e., mineral carbonation) capabilities, thereby enhancing its value in the context of CO_2 utilization. Our previous research demonstrated that ternesite possesses high carbonation reactivity and substantial CO_2-binding capacity, capable of sequestering 15.1% CO_2 within 2 h while achieving a compressive strength of up to 91.95 MPa [34]. During carbonation, ternesite can generate various forms of calcium carbonate (CC), including amorphous calcium carbonate (ACC), calcite, aragonite, and vaterite, as well as gypsum and silica gel (Si-gel) [37–39]. These carbonation products provide a combination of physical filling, chemical nucleation, and secondary hydration potential, collectively enhancing the cementitious properties of the system [40–44]. Accordingly, this dual characteristic of low calcium synthesis and high-efficiency carbonation endows ternesite with high applicability and added value in the field of cementitious materials. Furthermore, unlike other silicate minerals, the presence of the S element in ternesite may affect its carbonation behavior, especially the carbonation products, such as the crystalline morphology of CC and the production of gypsum [36,45]. However, the carbonation kinetics and mechanism of ternesite remain unclear. Most existing studies rely on gas-solid reaction systems, where limited mass-transfer efficiency restricts the accurate elucidation of dissolution-precipitation kinetics and product-evolution pathways. More importantly, a comprehensive understanding is still lacking regarding the role of sulfur in regulating CC crystal morphology, governing Si-gel leaching behavior, and controlling gypsum-formation mechanisms during carbonation. These knowledge gaps lead to limited controllability of the carbonation process, thereby constraining the design and performance optimization of carbonation products. Therefore, clarifying the carbonation mechanism of ternesite and elucidating the regulatory role of sulfur is essential.

Accordingly, this study employs a wet carbonation method to accelerate the carbonation of ternesite. The primary objectives are to elucidate the carbonation behavior, kinetics and mechanism of ternesite, and to provide a theoretical support for the application of carbonated ternesite as a carbonate binder or supplementary cementitious material (SCM). Initially, high-purity ternesite minerals were synthesized and subsequently subjected to carbonation for controlled reaction durations. The evolution of carbonation products was characterized using thermogravimetric (TG), X-ray diffraction (XRD), Fourier transform-infrared spectroscopy (FT-IR), and solid-state nuclear magnetic resonance (NMR). Additionally, the carbonation reaction process and kinetics of ternesite were elucidated by monitoring changes in carbonation heat, pH, and ion concentration. The microstructure and morphology of carbonated ternesite were examined using scanning electron microscopy (SEM), while surface characteristics associated with newly formed carbonation products were assessed via true density and Brunauer-Emmett-Teller (BET) analysis. Finally, the carbonation behavior of ternesite was systematically characterized, with particular emphasis on the evolution of polycrystalline CC, the leaching dynamics of active Si-gel, and the influence mechanism of SO_4^{2-}.

2. Experimental scheme

2.1. Synthesis of ternesite mineral

Analytical-grade chemical reagent powders, supplied by Sinopharm Chemical Reagent Co., Ltd., were used as raw materials in this experiment. $CaCO_3$, $CaCO_4{\cdot}2H_2O$, and SiO_2 were thoroughly mixed in a stoichiometric ratio (4: 2: 1). The homogenized powder was subsequently blended with 10 wt% deionized water and compacted into cakes with dimensions of Φ50 mm × 10 mm cakes using a stainless-steel mold at a pressure of 10 MPa. The raw meal cakes were subjected to solid-phase sintering in a high-temperature furnace, with the temperature ramped from room temperature to 1150 °C at a rate of 10 °C/min. The clinker was maintained at 1150 °C for 2 h to ensure mineral purity, then removed from the furnace and cooled using forced air. Subsequently, the cooled clinker was ground in a planetary ball mill for 30 min and stored for further use. To ensure experimental consistency, the above procedure was repeated multiple times to prepare ternesite powders, which were then combined and reserved. The synthesized ternesite was subjected to phase identification, particle size distribution, and morphology observation, with the results shown in Fig. 1. Quantitative analysis using the Rietveld refinement method indicated that the synthesized ternesite had a purity of 97.35%, with small amounts of larnite and anhydrite (Fig. 1(a)). The particle size distribution of ternesite powder is shown in Fig. 1(b), and the SEM image in Fig. 1(c) revealed rod-like crystal particles with dimensions around 2 μm. The true density of ternesite, as measured by a true density meter, was 2.99 g/cm^3.

2.2. Preparation of carbonated samples

Following successful synthesis, the ternesite mineral was subjected

Table 1
CaO content, CO_2 emission and formation temperature of clinker minerals.

Minerals	CaO content (%)	Formation heat *ΔH* (kJ/kg)	CO_2 emission from coal combustion (g/kg)	CO_2 emission from raw material decomposition (g/kg)	Total CO_2 emission (g/kg)	Formation temperature (°C)
C_3S	73.7	1856	285	578	863	1450
β-C_2S	65.1	1343	204	511	715	1300
γ-C_2S	65.1	1343	204	511	715	1200
C_3S_2	58.3	1115	169	458	627	1350
α-CS	48.3	839	127	379	506	1500
$C_5S_2\$$	58.3	1346	204	367	571	1150

This table was adapted from reference [34–36]

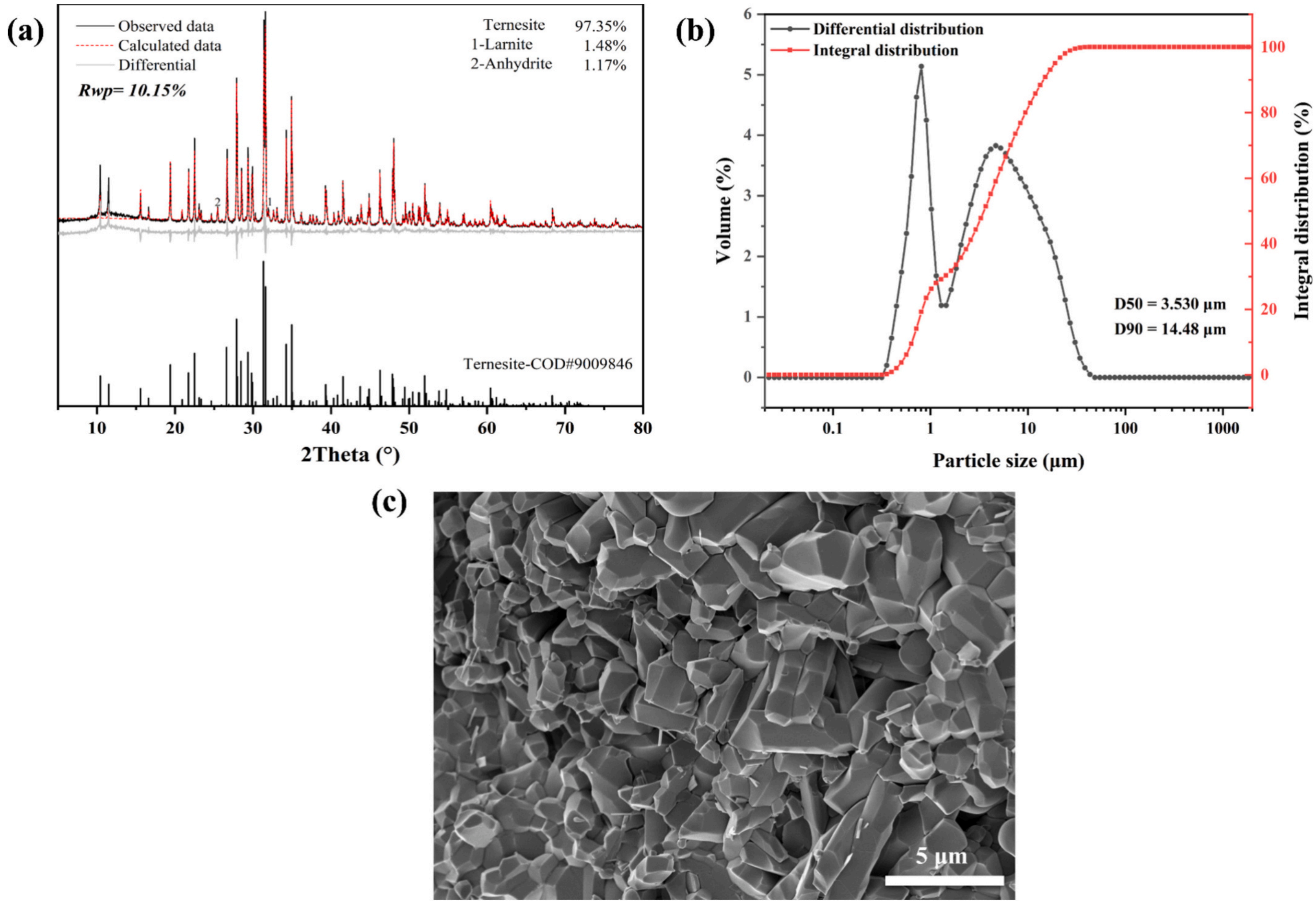

Fig. 1. Analysis of synthetic ternesite including phase identification, particle size distribution, and morphology: (a) XRD and Rietveld plots; (b) particle size distribution; (c) SEM image of the synthesized ternesite.

to wet carbonation treatment, with the detailed experimental procedure illustrated in Fig. 2. Ternesite powder was dispersed in a beaker containing water at a liquid-to-solid ratio of 10, with the water bath temperature maintained at 40 °C and the stirring speed set to 300 rpm. The suspension was pre-stirred for 1 min to ensure uniform dispersion of the powder. Subsequently, 99.9% CO_2 was introduced into the

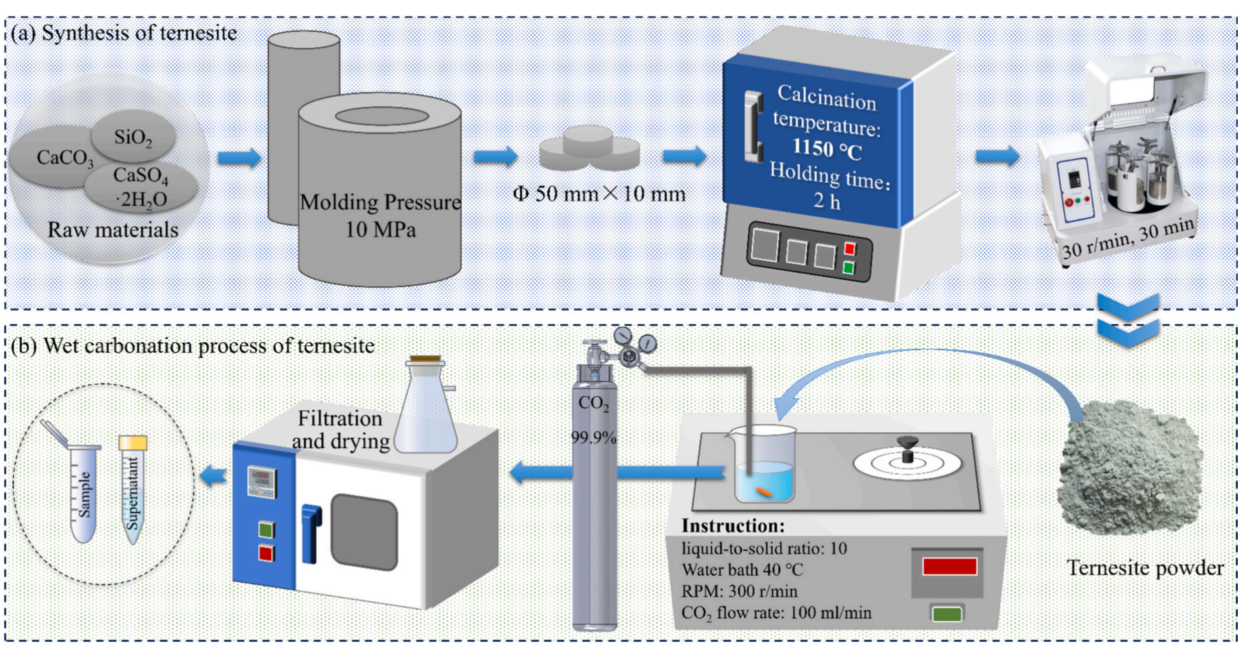

Fig. 2. Schematic diagram of the (a) synthesis and (b) wet carbonation of ternesite.

suspension at a flow rate of 100 mL/min using a gas flow meter. After the introduction of CO_2, the reaction was timed, with durations of 1 min, 2 min, 5 min, 10 min, 15 min, 30 min, 60 min, and 120 min. At the end of each reaction period, a vacuum filtration apparatus was employed to separate the solid residue from the filtrate, both of which were collected and stored separately. The filtrate was utilized for pH measurement and ion concentration analysis. The solid samples were dried in an oven at 40 °C for 24 h to remove free water, and were designated as WC-X, where X denotes the carbonation reaction time in minutes.

2.3. Test methods

2.3.1. Thermogravimetric analyzer

About 10 ± 1 mg of sample powder was placed in an alumina crucible and heated under a nitrogen atmosphere using a simultaneous thermal analyzer (Mettler-Toledo TGA/DSC1). The sample was heated from 50 °C to 1000 °C at a rate of 10 °C/min, and the TG curve was recorded. The first-order derivative of the TG curve was obtained using the STARe Evaluation Software to generate the DTG curve, which reflects the decomposition temperature and weight loss corresponding to the decomposition peaks [43]. The CO_2 sequestration capacity is the mass of CO_2 sequestered per kilogram of raw material, calculated as Eq. (1) [10]:

$$CO_2 \text{ sequestration (g/kg)} = \frac{CO_2 \text{ (wt.\%)}}{100 - CO_2 \text{ (wt.\%)}} \times 1000 \tag{1}$$

where CO_2 (wt.%) is the mass of decarbonation of the sample in the range of 400–850 °C.

Based on Eq. (1), the formula for carbonation degree can be derived as Eq. (2) [46]:

$$\delta_{CaO}\ (\%) = \frac{\frac{CO_2\ (\text{wt.\%})}{100 - CO_2\ (\text{wt.\%})} \times \frac{1}{Mw_{CO_2}\,(\text{g/mol})}}{CaO_{total}\ (\text{kg/kg})/Mw_{CaO}(\text{g/mol})} \times 100 \tag{2}$$

where Mw_{CO_2} and Mw_{CaO} represent the molar masses of CO_2 and CaO, respectively. CaO_{total} is the total CaO content in the raw material.

2.3.2. X-ray diffraction

An X-ray diffraction analyzer (XRD, Bruker D8 Advance) with a Cu target was used to test ternesite and carbonated samples. The scan range was 5–80°, with a step length of 0.02° and a scan time interval of 0.5 s. The EVA software was used for phase identification of the obtained XRD curves. Based on the Rietveld method, the TOPAS 4.2 software was utilized for quantitative analysis of samples containing an internal standard (15 wt% ZnO), with a fitting result credibility (Rwp < 15%) [10,43]. The entire region was fitted using the Scherrer formula for full-spectrum fitting, and a single peak was fitted for the local region. The crystallinity, crystal grain size, and unit cell parameters of different CC were calculated using the TOPAS 4.2 software. In addition, single-peak fitting of local regions was performed to obtain the crystallinity, crystallite size and cell parameters of the different crystalline phases based on the Scherrer formula [47]. The structural files used in this study were obtained from the Crystallography Open Database (COD), with the specific list provided in Table 2.

2.3.3. Fourier transform infrared spectroscopy

After mixing the samples with KBr powder at a ratio of 1:100 and pressing into tablets, Fourier transform infrared spectroscopy (FT-IR, Bruker EQUINOX 55) was used to detect infrared absorption spectra in the range of 400–4000 cm^{-1}.

2.3.4. Nuclear magnetic resonance

Nuclear magnetic resonance (NMR) experiments were conducted on an Agilent DD2 500 (B0 = 11.7 T) spectrometer to observe the Si—O tetrahedral structure in the samples. The ^{29}Si MAS-NMR spectrum used a 6 mm probe, with a resonance frequency of 99.29 MHz and a rotation speed of 4 kHz, employing high-power proton decoupling. Kaolin was used as the standard sample and 600 cumulative acquisitions were made during a 5 s relaxation time. The spectra were deconvolved using a Gaussian model. Additionally, ^{13}C {^{1}H} cross-polarization (CP) MAS NMR was employed in this study to detect carbonate anions in the samples.

Table 2
Structure files codes of minerals used for the Rietveld method.

Phase	Formula	COD Codes
Ternesite	$Ca_5(SiO_4)_2SO_4$	9,009,846
Larnite	Ca_2SiO_4	9,012,792
Gypsum	$CaSO_4·2H_2O$	5,000,039
Anhydrite	$CaSO_4$	5,000,040
Calcite	$CaCO_3$	9,000,095
Aragonite	$CaCO_3$	9,015,893
Vaterite	$CaCO_3$	9,016,215
Zincite	ZnO	9,004,180

2.3.5. Reaction heat

In order to accurately evaluate the carbonation reaction progress of ternesite, the carbonation reaction heat was measured by referencing the method of measuring hydration heat. The isothermal calorimeter (TAM Air) from TA Instruments recorded the reaction heat during the wet carbonation process of ternesite, with the ambient temperature maintained at 25 °C. A quantity of 5 g of ternesite powder was weighed and transferred into an ampoule, followed by the addition of 25 g of deionized water. The ampoule was then placed within the instrument channel and left to stabilize for a period of 45 min. Thereafter, the motor was initiated to facilitate internal stirring, concurrently with the introduction of CO_2. The motor speed was set at 15 rpm, the CO_2 partial pressure at 1 atm and the flow rate of 100 mL/min. To ensure heat conservation, the reactor was equipped with sealing rings and insulation wool, and the entire system was placed in a temperature-controlled laboratory. Additionally, a 30-min baseline recording was maintained after reaction completion to confirm the absence of systematic thermal drift.

2.3.6. pH value and ion concentration

A pH meter (PHS-3C) from Shanghai LeiCi Company was used to record the changes in pH value of the solution during the carbonation reaction. The filtered filtrate was mixed evenly with 6.5% nitric acid at a ratio of 1:10, and the ion concentration was detected using an inductively coupled plasma-optical emission spectrometer (PerkinElmer Avio 500 ICP-OES).

2.3.7. True density

The true density of the dried sample powder was tested using a true density meter (JW-M100A). The instrument was pre-treated five times before formal measurement, and the final result was the average of five measurements.

2.3.8. Scanning electron microscope

A scanning electron microscope (SEM, HITACHI, SU5000) was used to observe the micro-morphology and particle size of the samples. Additionally, samples embedded in resin were observed in backscattered electron (BSE) mode, and energy dispersive spectrometer (EDS) was used to collect the elemental composition and distribution of the samples. Before imaging, the sample surface was sputtered with gold for 90 s to enhance conductivity.

2.3.9. Brunauer-Emmett-Teller analysis

The Brunauer-Emmett-Teller (BET) N_2 adsorption-isotherms were recorded using a surface area and pore diameter analyzer (ASAP 2460) from Micromeritics, USA, which used nitrogen as the adsorbent to measure the amount of nitrogen adsorbed by the samples at different pressure points.

3. Results and analysis

3.1. CO_2 sequestration capacity and carbonation degree of ternesite

3.1.1. TG analysis

The TG-DTG results (Fig. 3) show that the samples exhibited two prominent weight-loss stages, corresponding to dehydration and decarbonation. The weight loss observed during heating from 50 °C to 300 °C was primarily attributed to the dehydration of phases such as calcium silicate hydrate (C-S-H), Si-gel, and gypsum [33]. The decarbonation of the samples mainly occurred between 400 °C and 850 °C, which included the decarbonation of ACC and poorly crystalline calcium carbonate (PCCC) at 400–620 °C, as well as the decomposition of crystalline CC within the 620–850 °C range [47,48].

A broad dehydration-related weight loss peak was observed on the DTG curve between 100 °C and 175 °C, with a particularly pronounced intensity at 10 min, indicating an elevated formation of hydroxyl-containing substances such as gypsum and Si-gel. Additionally, a distinct weight loss peak associated with CC decomposition appeared within the lower-temperature region of 400–620 °C. As carbonation progressed, the extent of decarbonation within this temperature interval first increased and then declined sharply after 10 min, reflecting the transient formation and subsequent transformation of ACC or PCCC [49,50]. During carbonation, ACC acts as a metastable precursor that gradually transforms into crystalline CC [51]. As the reaction continues, ACC progressively converts into the thermodynamically more stable CC polymorphs. Therefore, the weight loss initially increased and then decreased at lower decomposition temperatures (450–620 °C). Meanwhile, the weight loss peak of crystalline CC increased as the reaction progressed. By 30 min, the mass loss of the carbonated samples approached a steady state, indicating that carbonation had effectively reached completion and that the CC content had plateaued. Moreover, the decomposition temperature peak of CC shifted progressively to higher values with increasing reaction time, but dropped slightly after 30 min of carbonation. These variations in decomposition temperature primarily reflect changes in CC crystallinity, particle size, and polymorphic structure [47,49]. Among the CC polymorphs, calcite exhibits the highest thermodynamic stability, whereas aragonite and vaterite remain metastable and decompose at comparatively lower temperatures. The observed decrease in decomposition temperature indicates a shift in CC polymorphism, suggesting a reduction in calcite content accompanied by increased proportions of aragonite and vaterite.

3.1.2. Carbonation reaction process

Fig. 4 illustrates the evolution of CO_2 sequestration capacity and carbonation degree of ternesite with increasing reaction time. After 5 min of CO_2 introduction, the carbonation degree of ternesite reached only 21.4%. When the reaction reached 10 min, the carbonation degree reached 75.2%, corresponding to a CO_2 sequestration capacity of 268.4 g/kg. The carbonation process was essentially complete after 15 min, with the carbonation degree exceeding 95%. After 30 min, the carbonation degree was 95.8%, corresponding to a CO_2 sequestration capacity of 342.0 g/kg. The maximum carbonation degree of 96.1% and CO_2 sequestration capacity of 342.8 g/kg was achieved after 120 min of reaction. Notably, the carbonation degree exhibited a slight fluctuation at 60 min, reaching 95.2%.

The variation in the carbonation rate was determined by calculating the first derivative of the carbonation degree, as depicted in the inset of Fig. 4. The reaction rate decreased rapidly within the first 2 min, increased significantly between 2 and 10 min to reach its maximum, then slowed down between 10 and 30 min, and subsequently stabilized. The carbonation reaction is a typical acid-base neutralization process, characterized by significant exothermicity during the initial stage [52]. According to Henry's law, the solubility of CO_2 decreases with increasing temperature. The low solubility of ternesite in water and the limited amount of free Ca^{2+} in the solution led to a rapid decrease in the reaction rate within the first 2 min. Thus, the solubility of CO_2 and the degree of hydrolysis of ternesite limited the initial carbonation reaction rate [45]. As CO_2 continued to be introduced, the decreasing solution pH promoted ternesite dissolution, thereby significantly increasing the early reaction rate. After 10 min of carbonation, ternesite was essentially completely dissolved, resulting in the formation of a large amount of CC (ACC, PCCC, and polycrystalline CC), gypsum, and Si-gel. After 10 min

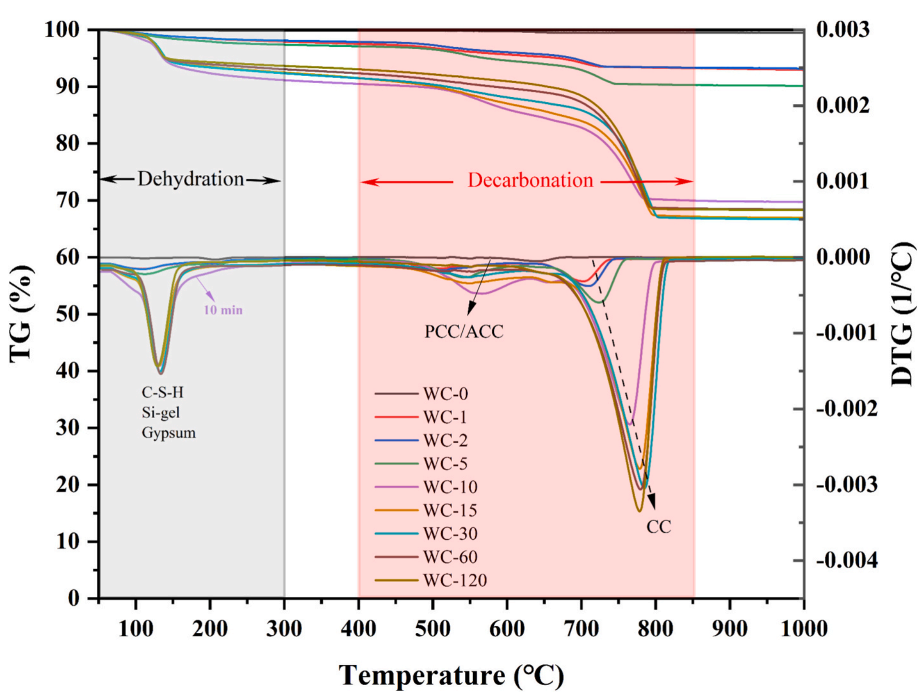

Fig. 3. TG-DTG curves of ternesite at different times of wet carbonation.

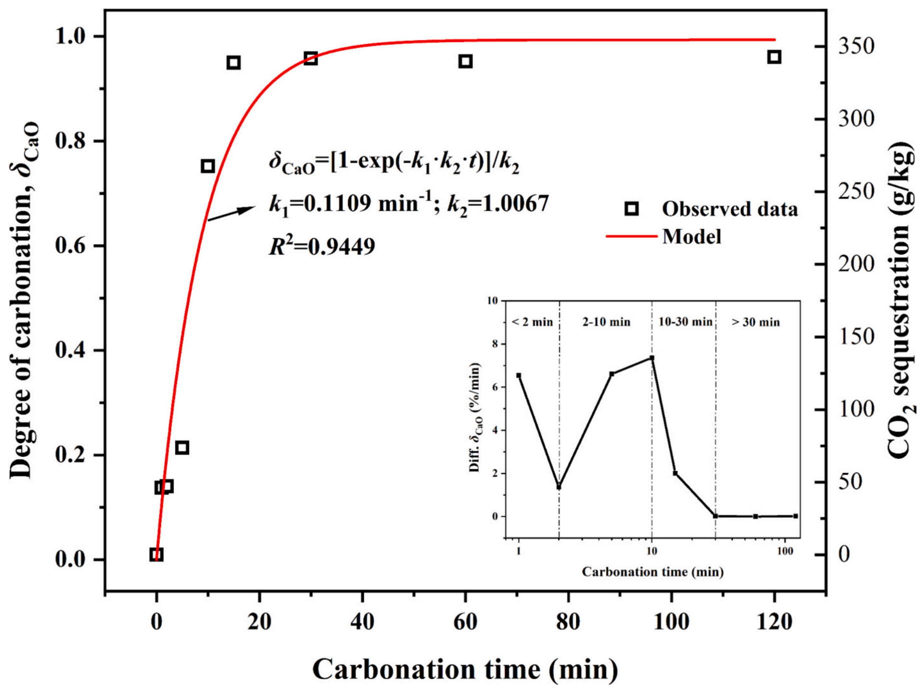

Fig. 4. Carbonation degree, kinetics, and CO_2 sequestration capacity for ternesite.

of carbonation, the carbonation rate gradually decreased, attributed to the decreasing pH and the formation of a CC layer [53,54]. After 30 min, the carbonation reaction essentially reached equilibrium.

A surface-coverage model was adopted to describe the kinetics of the ternesite carbonation reaction, as outlined in Eqs. (3) to (7) [55,56]. In these equations, Φ^n represents the function of product deposition; S_g (m^2/g) denotes the specific surface area of the reactant; M (g/mol) is the molar mass; k_s ($mol/min/m^2$) is the overall reaction rate constant, which depends on the dissolution rate of CO_2, the leaching rate of Ca^{2+}, and the precipitation rate of carbonates; k_p (m^2/mol) is a proportionality constant, which is influenced by reaction temperature, relative humidity, and reactant concentration, and reflects the proportion of the reactant surface not covered by the product. By substituting Eqs. (5) and (6) into Eqs. (3) and (4) and integrating, the functional relationship between carbonation degree and time can be established. Typically, $n = 1$ is chosen for analyzing the kinetics of wet carbonation, as shown in Eq. (7). Here, k_1 (min^{-1}) is the overall reaction rate constant, and k_2 is the proportionality constant. The fitting curve is shown in Fig. 4. The curve fitting yielded $k_1 = 0.1109\ min^{-1}$ and $k_2 = 1.0067$, with a variance $R^2 = 0.9449$ between the measured values and the fitted data. These results indicate that the model provides a reliable fit and can accurately predict the carbonation reaction process of ternesite.

$$d\delta_{CaO}/dt = S_g \bullet M \bullet k_s \bullet \Phi \tag{3}$$

$$-d\Phi/dt = k_p k_s \Phi^n \tag{4}$$

$$k_1 = k_s \bullet S_g \bullet \mathrm{M} \tag{5}$$

$$k_2 = k_p/(S_g \bullet \mathrm{M}) \tag{6}$$

$$d\delta_{CaO} = [1 - \exp(-k_1 \bullet k_2 \bullet \mathrm{t})]/k_2, \ \ for\ n = 1 \tag{7}$$

3.2. Carbonation kinetics of ternesite

3.2.1. Phase evolution

The XRD patterns of ternesite after wet carbonation at different times are shown in Fig. 5. As the carbonation reaction proceeded, the intensity of the ternesite diffraction peaks gradually diminished, while new diffraction peaks corresponding to calcite emerged. After 10 min of CO_2 introduction, the diffraction peaks of the ternesite disappeared entirely, while those of aragonite and gypsum increased markedly, and a weak diffraction peak attributable to vaterite was also observed. After 120

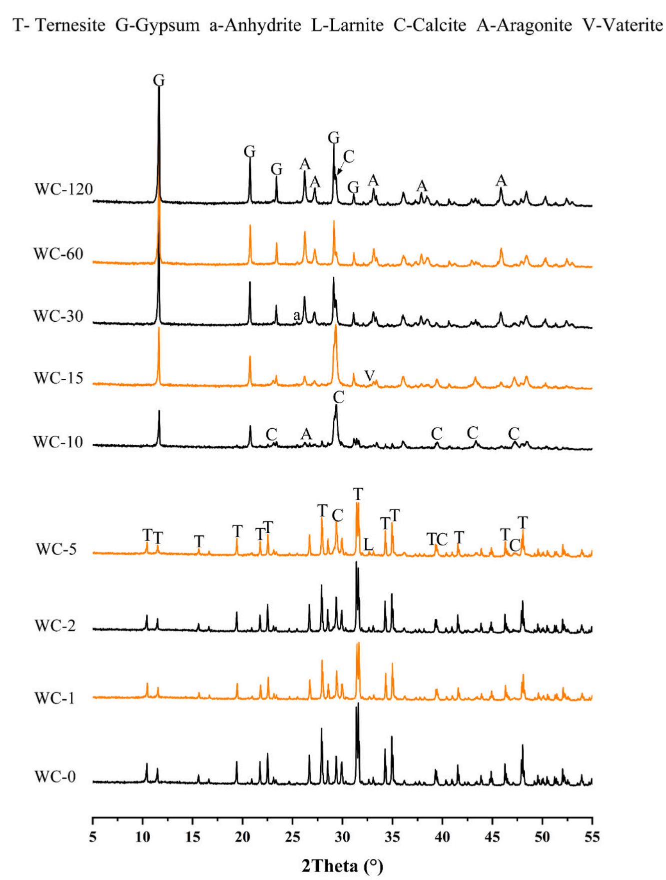

Fig. 5. XRD patterns of ternesite at different times of wet carbonation.

min of reaction, the wet carbonation products of ternesite mainly included different crystal forms of CC, gypsum, and amorphous phases.

Quantitative analysis based on the XRD patterns was performed to determine the phase composition of samples at different carbonation times, with the calculations exhibiting satisfactory fitting ($Rwp < 15\%$). As shown in Fig. 6, the content of ternesite decreased continuously as carbonation progressed, whereas the content of CC, gypsum, and amorphous phase increased. After 10 min of reaction, the ternesite content sharply declined from 79 wt% to 12 wt%, the amorphous phase content increased from 6 wt% to 40 wt%, and the amounts of calcite, aragonite, and gypsum also rose significantly. These results indicate that the carbonation reaction was particularly intense at 10 min, leading to the breakdown of the original ternesite crystal structure and the formation of substantial quantities of carbonation products. Notably, the amorphous phase content decreased between 10 and 30 min, while the content of CC (aragonite and calcite) increased markedly. This phenomenon can be attributed to the transformation of ACC within the amorphous phase via a dissolution-reprecipitation process into more stable crystalline forms, resulting in a gradual increase in polycrystalline CC [50], consistent with the previous TG analysis. After 30 min, the carbonation reaction was essentially complete, yielding final products comprising approximately 50 wt% CC, 30 wt% amorphous phases, and about 16 wt% gypsum. Subsequent FT-IR and NMR analyses suggest that the amorphous phase is primarily composed of highly reactive Si-gel.

Notably, the CC content of different crystalline forms in the product changed as the reaction proceeded. The calcite content peaked at 30 wt % after 15 min of carbonation, then declined rapidly by 30 min. Conversely, aragonite content increased steadily throughout the reaction, becoming the predominant carbonate phase and reaching a maximum of 45 wt% at 60 min. These trends may be attributed to the crystal structure and solution chemistry of ternesite [57]. In addition, ternesite produced a certain percentage of gypsum after carbonation, and the existence of sulfate may affect the growth and morphology of CC [45]. After 60 min of carbonation, the calcite content increased while aragonite content decreased, reflecting the greater thermodynamic stability of calcite compared to aragonite [49].

Fig. 7 presents the FT-IR spectra of ternesite after wet carbonation treatment at different times, while Table 3 summarizes the wavenumbers corresponding to the distinct characteristic peaks and vibrational bands. The FT-IR spectra exhibit pronounced characteristic peaks for ternesite, particularly those associated with Si—O vibrations at

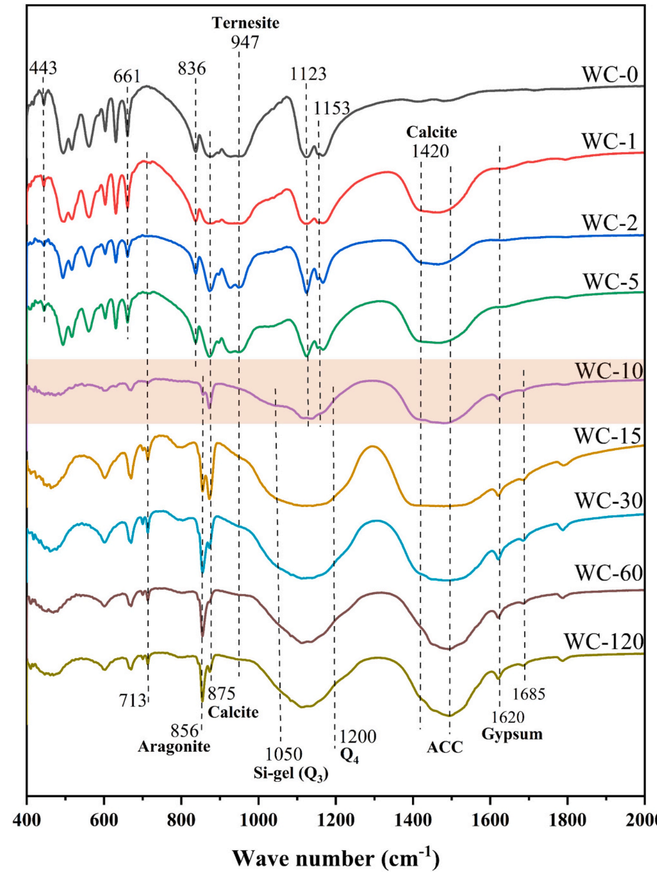

Fig. 7. FT-IR spectra of ternesite at different times of wet carbonation.

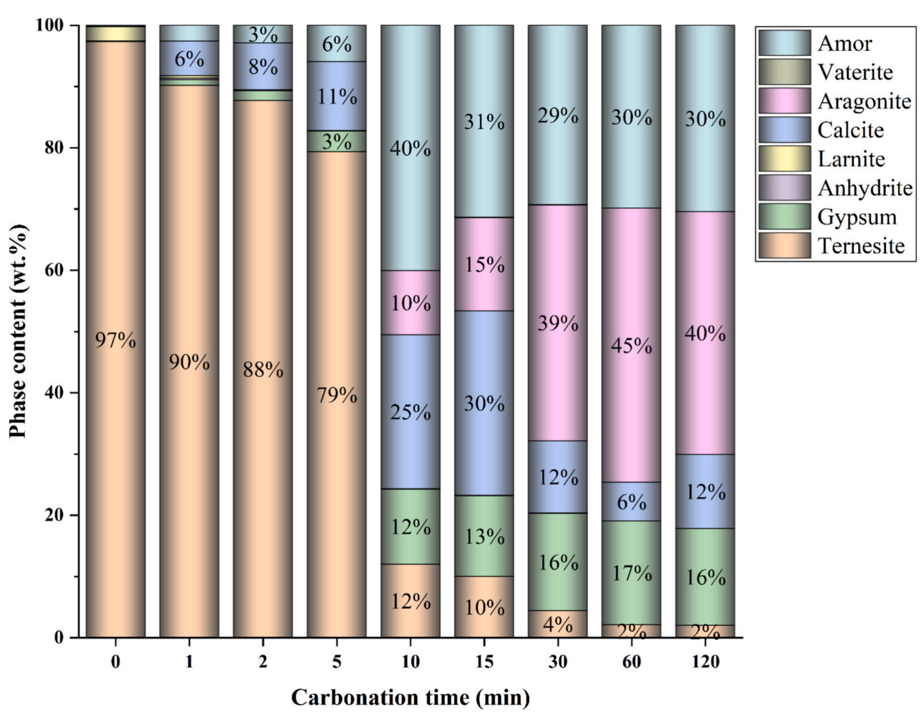

Fig. 6. Phase composition of ternesite at different times of wet carbonation.

Table 3
Typical wavenumbers of FT-IR absorption.

Phase	Group	Absorption band (cm^{-1})				Ref.
		Symmetric stretching (v_1)	In-plane bending (v_2)	Asymmetric stretching (v_3)	Out of plane bending (v_4)	
$C_5S_2\$$	Si-O	836	443	947	491	[33,34]
	S-O	–	412, 516	1123, 1153	558, 602, 630, 661	
Calcite	C-O	1080	875	1420	713	[34,41,42]
Aragonite	C-O	1083	856	1450, 1470	713	[34,42]
Vaterite	C-O	1087	875	1450–1490	744	[41]
ACC	C-O	1074	866	1420–1480	–	[50,59]
$CaSO_4·2H_2O$	S-O	–	412, 516	1120, 1153	558, 602, 630, 661	[33,34,37]
	H-O-H	–	1620, 1685	–	–	
Si-gel	Q^3	–	–	1050–1080,	–	[34,50,58]
	Q^4	–	–	1170–1250	–	

443 cm^{-1} (v_2), 491 cm^{-1} (v_4), 836 cm^{-1} (v_1), and 947 cm^{-1} (v_3). As the carbonation reaction proceeded, the intensity of the calcite characteristic peak at 875 cm^{-1} (v_2) increased progressively. After 10 min of carbonation, the characteristic peaks of ternesite were no longer observed, while a new peak corresponding to aragonite emerged at 856 cm^{-1} (v_2). Broad absorption bands in the range of 1050–1200 cm^{-1} and 1420–1480 cm^{-1} are indicative of the presence of Si-gel and ACC. Additionally, the characteristic peak for the stretching vibration of gypsum appeared at 1620 cm^{-1}. Notably, the vibrational absorption bands associated with ACC diminished progressively during carbonation, whereas the intensities of the bands at 856 cm^{-1} and 875 cm^{-1} increased, suggesting the transformation of ACC into crystalline CC. Throughout the carbonation process, ACC can be considered as a precursor to the formation of crystalline CC, typically undergoing dissolution-re-precipitation and transforming into calcite at lower temperatures or aragonite at higher temperatures [58].

The ^{29}Si MAS NMR spectra of the carbonate samples are shown in Fig. 8, including measured data and deconvolution results for some samples. The deconvolution of the NMR spectra enables the identification of silicon site environments within the silicate tetrahedra and allows for the calculation of their relative intensity ratios. Q^n sites ($n = 1$, 2, 3, 4) indicate the number of oxygen atoms shared by each silicate tetrahedron with its neighboring tetrahedra. Q^0 sites (approximately −74 ppm) correspond to monomeric silicate tetrahedra, which are typically present in C_2S [47]. In C-S-H, there are chain-like silicate tetrahedra, with terminal silicate tetrahedra Q^1 (−76.8 ppm to −79.9 ppm) and dimers Q^2 (Q^{b2}, −82.4 ppm bridging site; Q^{p2}, −84.6 ppm, paired sites) [60,61]. As the carbonation reaction proceeds, the silicon

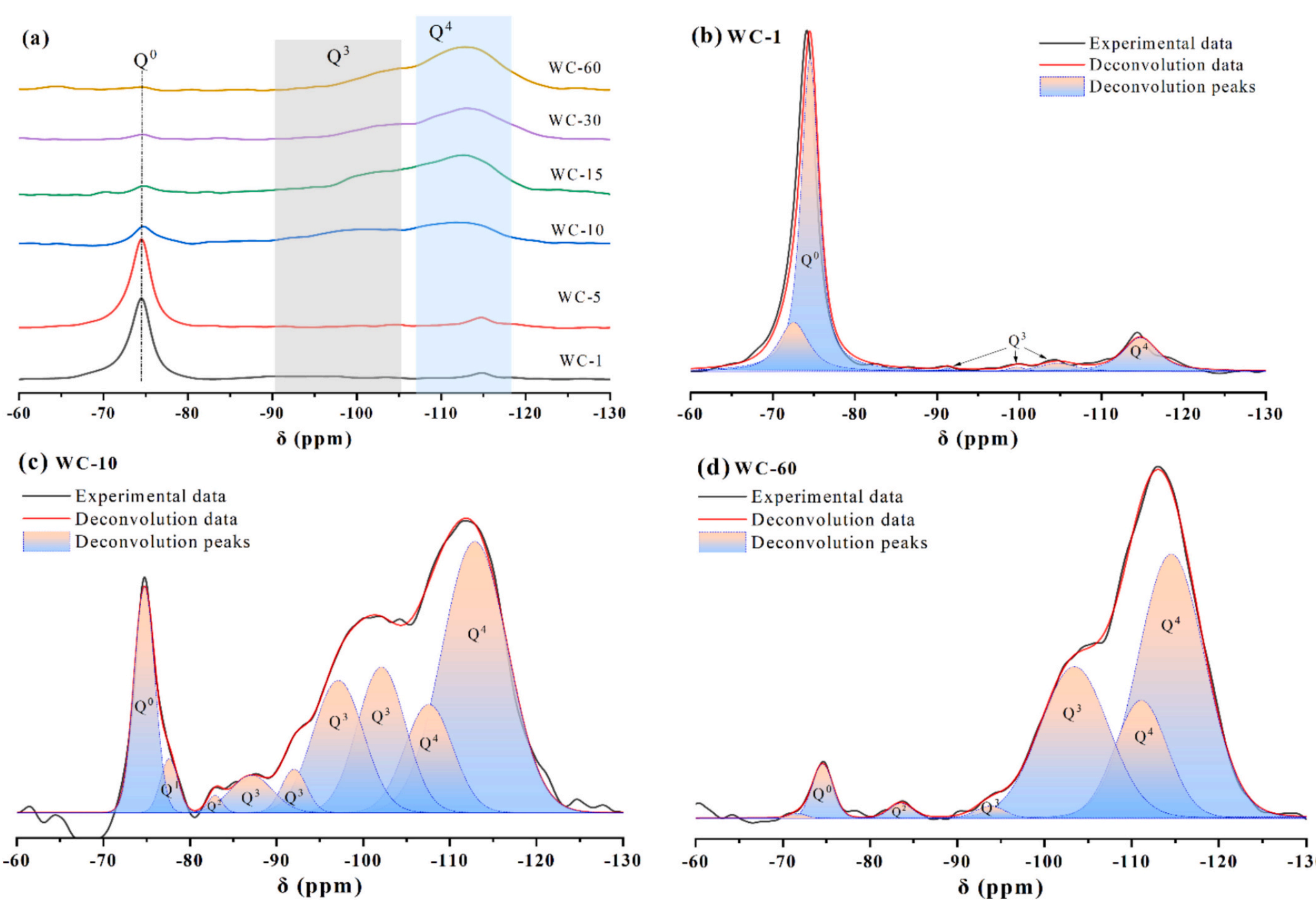

Fig. 8. (a) ^{29}Si MAS NMR spectra of ternesite at different times of wet carbonation; (b) deconvolution results of WC-1; (c) deconvolution results of WC-10; (b) deconvolution results of WC-60.

chemical environment undergoes significant changes. The occurrence of Q^3 (−89.7 ppm to −104.8 ppm) and Q^4 (approximately −114 ppm) sites indicates the formation of silicate tetrahedra with double-chain, layered structures or three-dimensional networks, reflecting the decomposition of C-S-H or polymerization of Si-gel during the carbonation process [34,50,58].

Table 4 shows the relative intensity ratios of silicate tetrahedral sites, where the sum of Q^1 and Q^2 represents the relative proportion of C-S-H, and the sum of Q^3 and Q^4 represents the relative proportion of Si-gel [58]. Q^1 and Q^2 signals emerged after 10 min of reaction, but their intensities were minimal and ultimately vanished. This does not accurately indicate the formation of C-S-H, and Q^1 and Q^2 may originate from highly decalcified ternesite or calcium-containing Si-gel [34,45]. The intensities of Q^3 and Q^4 significantly increased after 10 min of reaction, with more than 80% of Si existing in the form of Si-gel. From 15 min onwards, the relative proportion of Q^3 and Q^4 exceeded 95%, indicating that the Si—O in ternesite continues to polymerize inwards [45], reflecting the degree of carbonation reaction. Combining the analysis of carbonation products in the previous sections, the carbonation reaction of ternesite can be represented by Eq. (8) [34]. Additionally, whether the hydration reaction of ternesite as shown in Eq. (9) [33] occurs during the wet carbonation process remains in question. This uncertainty arises because calcium hydroxide (CH) was not detected by XRD or TG, and the intensities of Q^1 and Q^2 remained low. Even if a small amount of CH and C-S-H were formed, the hydration reaction of ternesite was very weak, and the intermediate phases would subsequently participate in the carbonation reaction to transform into CC and Si-gel [58]. Therefore, ternesite may be more suitable as a CO_2 mineralization material rather than as a hydration cementitious material, and may also serve as an effective precursor for the preparation of high-purity, highly reactive Si-gel.

$$C_5S_2\$ + 4H\overline{C} \rightarrow 4C\overline{C} + C\$H_2 + 2SHn \tag{8}$$

$$C_5S_2\$ + 6H \rightarrow 2CH + C\$H_2 + 2C-S-H \tag{9}$$

Fig. 9 shows the $^{13}C\{^1H\}$ cross-polarization (CP) MAS NMR spectra of ternesite for carbonation at 10 min and 30 min. Since anhydrous carbonate phases exhibit no signals in the spectrum, only hydrated or HCO_3^- containing carbonates such as ACC and monohydrocalcite (MHC) show single peaks in the $^{13}C\{^1H\}$ CP/MAS NMR spectrum [62,63]. Therefore, the $^{13}C\{^1H\}$ CP/MAS NMR can be used to distinguish carbonate ions present in the anhydrous phase from those in aqueous or hydroxyl-containing environments [64]. As shown in Fig. 9, the sample carbonated for 10 min showed a signal peak near 168.9 ppm, indicating the formation of hydrated CC during the reaction. However, no diffraction peaks for crystalline hydro-carbonate were detected in the XRD curve. In addition, the full width at half maximum (FWHM) of the signal peak at 168.9 ppm in Fig. 9(a) is 3.0 ppm, consistent with previous reports [65]. Therefore, the signal observed in the $^{13}C\{^1H\}$ CP/MAS NMR spectrum can be attributed to the presence of ACC in the carbonated sample. As the reaction progressed, unstable ACC converted to CC crystals. Consequently, no distinct signal peaks are observed in the $^{13}C\{^1H\}$ CP/MAS NMR spectrum of WC-30. This finding coincided with prior TG and FT-IR analysis, reflecting the presence and content changes of ACC within the system.

Table 4
^{29}Si MAS NMR intensity of ternesite at different times of wet carbonation.

Samples	I(Q^0) (%)	I(Q^1) (%)	I(Q^2) (%)	I(Q^3) (%)	I(Q^4) (%)	I(C-S-H)[a] (%)	I(Si-gel)[b] (%)
WC-1	82.1	–	–	10.4	7.5	–	17.9
WC-5	81.5	–	–	4.9	13.6	–	18.5
WC-10	10.3	2.1	0.5	36.2	50.9	2.6	87.1
WC-15	3.5	1.0	0.3	14.5	80.7	1.3	95.2
WC-30	3.2	–	1.0	29.7	66.1	1.0	95.8
WC-60	4.5	–	–	23.2	72.3	–	95.5

[a] Calculated from the sum of I(Q^1) and I(Q^2);
[b] Calculated from the sum of I(Q^3) and I(Q^4).

3.2.2. Carbonation reaction heat

In order to meticulously evaluate the carbonation reaction kinetics of ternesite and accurately quantify its reaction rate, the dynamic heat evolution during the carbonation of ternesite was investigated using isothermal calorimetry, analogous to methods employed in hydration heat studies [47], the corresponding results are presented in Fig. 10. The carbonation process of ternesite proceeded through three distinct stages: (I) An acceleration stage, characterized by a peak in heat flow at approximately 7 min; (II) a deceleration stage, from 7 to 25 min; and (III) a stabilization stage, commencing after 25 min of reaction. The maximum reaction intensity was observed at 7 min, with a peak heat flow of 88 mW/g and a cumulative heat release of 67 J/g over 60 min. The observed trends in heat flow and cumulative heat are consistent with the carbonation reaction process shown in Fig. 4.

3.2.3. Solution chemistry change

The carbonation reaction is a typical acid-base neutralization process, in which the diffusion of Ca determines the carbonation rate, and elevated a high pH promotes carbonation process [66]. Analysis of pH variations and elemental concentrations (Ca, Si) during carbonation enables elucidation of the reaction mechanism of ternesite and allows for the establishment of quantitative relationships among mineral transformation, carbonation degree, and solution chemistry. The changes in pH value and ion concentrations during the wet carbonation of ternesite is shown in Fig. 11(a). Additionally, Fig. 11(b) shows the relationship between the carbonation degree of ternesite and pH and ion concentrations.

The initial pH value of ternesite in water was 11.42, attributable to the hydrolysis of the alkaline mineral. Upon introduction of CO_2 for 1–5 min, the solution pH value decreased and fluctuated around 9–10. This stage was primarily controlled by CO_2 dissolution and was also related to the degree of destruction of the ternesite crystal structure and ionic bonds [47]. The subsequent reaction kinetics were dictated by the precipitation rate of the carbonation products. At 10 min, ternesite was essentially dissolved, as evidenced by the absence of characteristic XRD diffraction peaks. After 15 min of CO_2 exposure, the carbonation degree of ternesite exceeded 90%, and the solution pH rapidly decreased from 11.42 to 7.32. These results indicate that the carbonation reaction of ternesite proceeds rapidly and predominantly under alkaline conditions. Carbonation products formed under alkaline conditions tend to form metastable CC [58,67], which partially accounts for the observed increase in aragonite content after 15 min of carbonation. Subsequently, during the remaining carbonation period, the pH stabilized at approximately 7, and the reaction rate gradually plateaued. This suggests that the CO_2 concentration in solution reached saturation, and ionic concentrations approached equilibrium [50].

In contrast to other alkaline minerals (C_3S, C_2S, CH, etc.) [10,47,56], the Ca concentration in ternesite increased rapidly during the initial stage of carbonation, reaching a maximum value at 5 min. Between 5 and 15 min, the Ca concentration decreased markedly, subsequently reaching equilibrium. According to Le Chatelier's principle [68], a decrease in solution alkalinity shifts the dissolution equilibrium of ternesite, promoting further mineral dissolution and increased Ca release. Meanwhile, CO_2 introduction facilitates the precipitation of CC. Therefore, after CO_2 was introduced, the Ca concentration initially increased (due to enhanced ternesite dissolution) and subsequently decreased as CC precipitation occurred. The final Ca concentration is determined by the dynamic balance between the dissolution and precipitation rates of calcium-bearing phases (CC and gypsum) [58].

The initial Si concentration was mineral, constrained by the limited hydration activity of the ternesite. After CO_2 introduction, the Si concentration rapidly increased, mainly due to accelerated ternesite

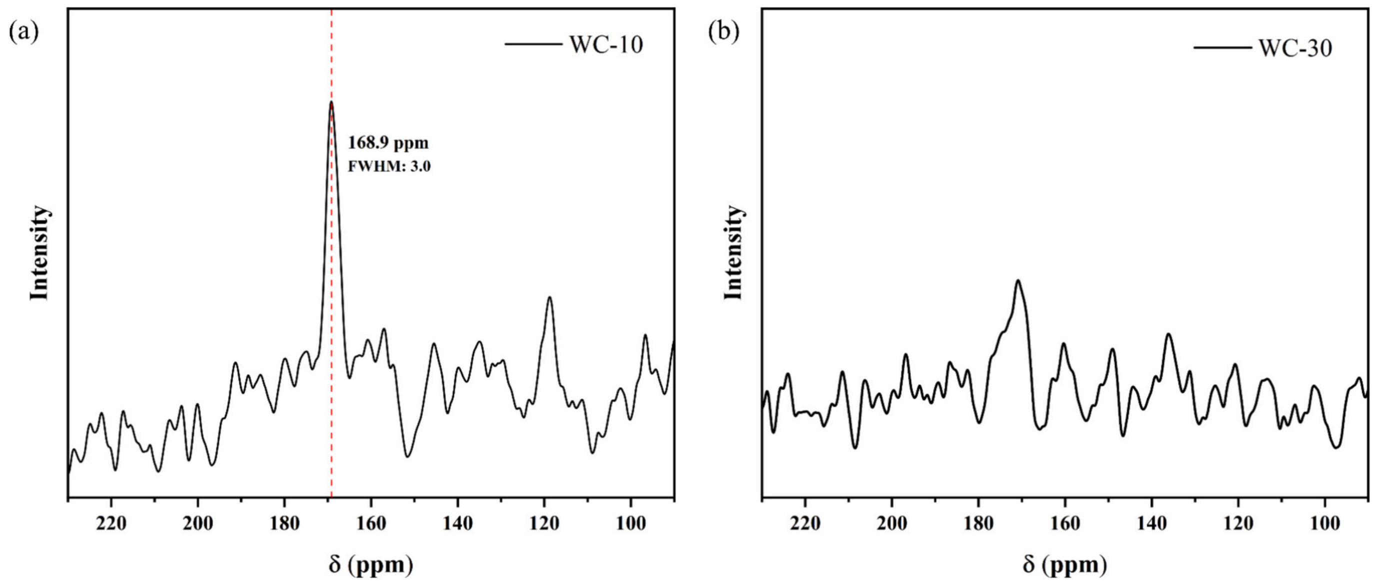

Fig. 9. $^{13}C\{^{1}H\}$ CP MAS NMR spectra of the carbonated ternesite.

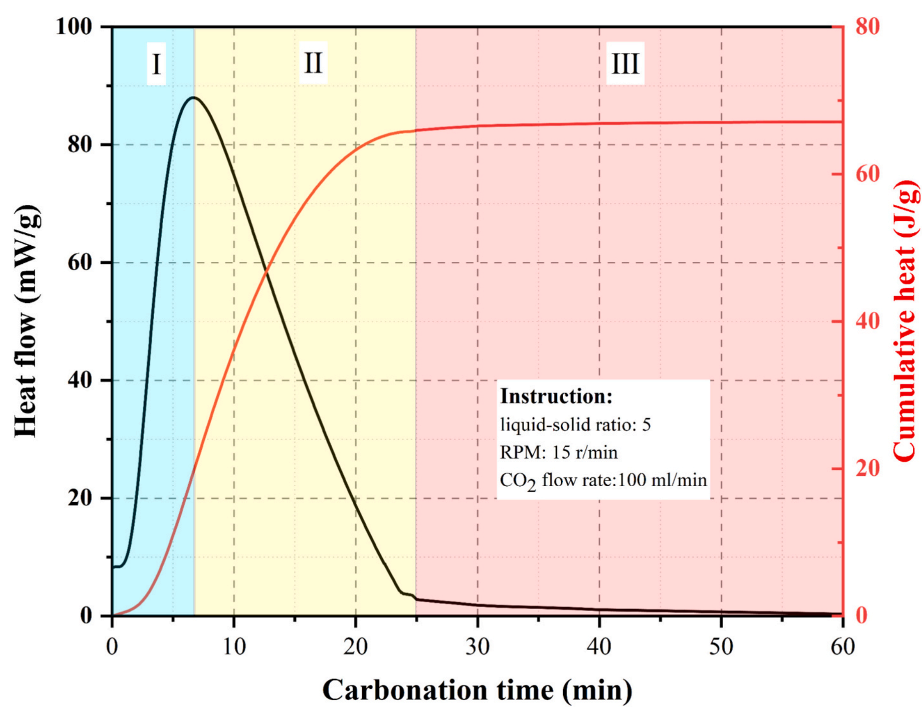

Fig. 10. Heat flow and total heat curves for ternesite carbonation reaction.

dissolution and subsequent Si leaching. Subsequently the Si—O in the system polymerized to form Si-gel, concurrent with carbonate precipitation. As shown in Section 3.2.1, after 15 min, more than 95% Si was incorporated into the gel phase and the Si-gel was essentially fully formed. During this process, a "core-shell" structure developed, comprising an outer CC shell, an intermediate Si-rich layer, and unreacted core particles [34,45]. After 15 min of reaction, the Si concentration continued to rise, coinciding with a carbonation degree exceeding 95% and near-complete formation of carbonation products. It was observed that there was a good correspondence between the Si concentration and the amount of aragonite. At 60 min, the maximum Si concentration corresponded to the highest aragonite content in the carbonation products. It is well established that, compared to the densely packed blocky calcite, needle-like aragonite exhibits greater porosity [42]. The increased aragonite content enhances the porosity of the "shell" structure, thereby facilitating the leaching of internal Si, a phenomenon that was verified in the subsequent microstructure analysis.

3.3. Microstructural development and phase distribution

3.3.1. Micro-morphology

Fig. 12 shows the microstructural evolution of ternesite and its carbonation products at different stages of the carbonation reaction. In aqueous solution, ternesite retained its characteristic rod-like morphology, exhibiting no significant alterations (Fig. 12(a)).

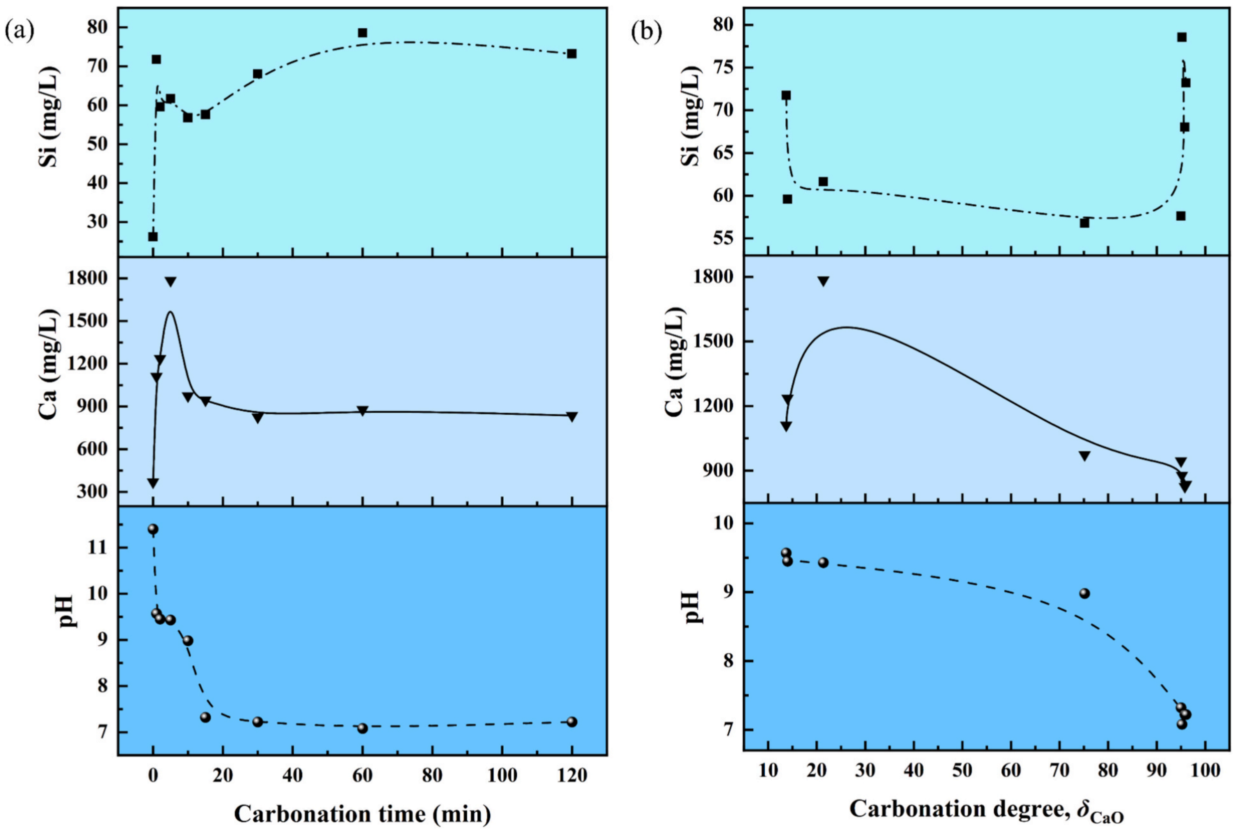

Fig. 11. (a) Solution pH, Ca and Si concentration changes and (b) relation with carbonation degree during wet carbonation of ternesite.

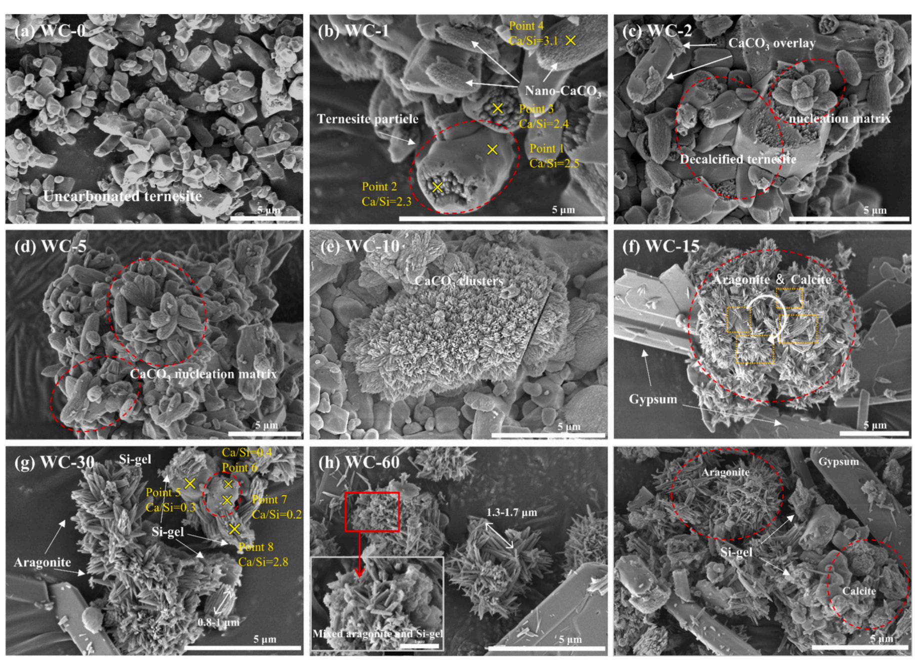

Fig. 12. SEM images and EDS results of ternesite at different times of wet carbonation.

Following CO_2 introduction, two distinct morphologies were observed (Fig. 12(b)): (1) ternesite particles exhibiting surface defects (point 1), and (2) elliptical matrices with surfaces coated by nano-CC particles (point 4). As the carbonation reaction proceeded, the ternesite underwent accelerated dissolution, releasing Ca^{2+}, SO_4^{2-}, OH^-, and H_4SiO_4, which resulted in the formation of surface pores and defects (Points 2, 3) [47]. As the concentrations of Ca^{2+} and CO_3^{2-} in solution increased, highly supersaturated CC began to precipitate, with nano-CC aggregating and growing on elliptical matrices measuring 1–2 μm in size. The nano-CC particles exhibited spherical or flake-like morphologies, thereby increasing the specific surface area of the matrix and providing additional nucleation sites for further CC precipitation [56]. This observation was confirmed in the subsequent BET analysis. With prolonged reaction time, more pronounced defects were observed on the ternesite particles, and the surface was covered with a distinct CC layer and nucleation matrix (Fig. 12(c)). Moreover, CC continued to precipitate and crystallize, aggregating and growing on the elliptical matrix. Along with the growth of CC crystal particles, the contour of the elliptical matrix surface gradually became distinct (Fig. 12(d)).

After 10 min of reaction, flower-clustered CC matrix was observed in Fig. 12(e), and intact ternesite particles largely disappeared. In combination with the contents of Section 3.2, at this point ternesite had been dissolved in large quantities, and CC, gypsum, and amorphous phases (ACC, PCCC, and Si-gel) were predominantly present in the system. ACC can be regarded as a precursor to the formation of polycrystalline CC (such as calcite and aragonite) [50]. During the dissolution-re-precipitation process of CC, accompanied by the reorganization of crystal structure and the increase in molar volume, the nucleation

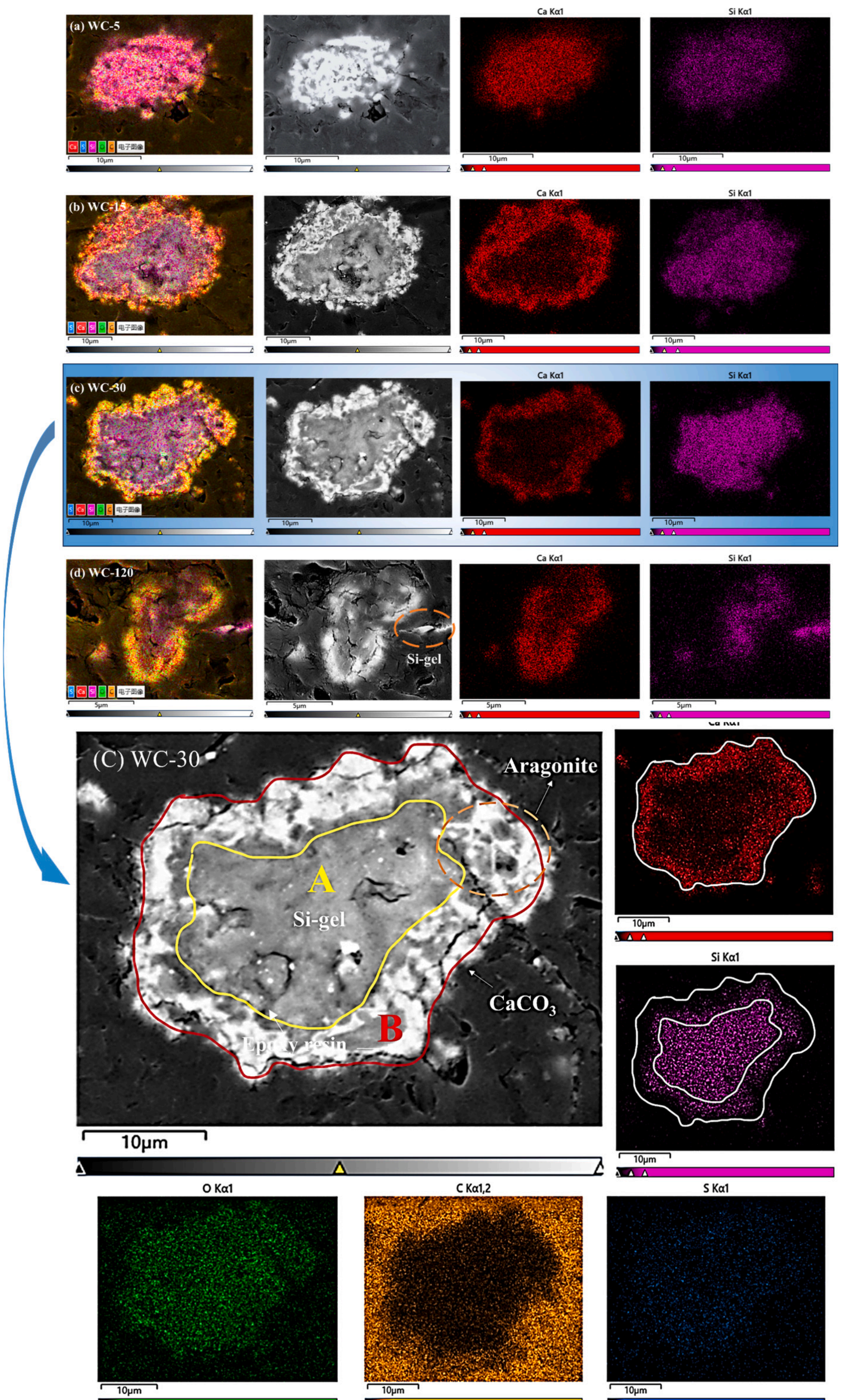

Fig. 13. BSE images and energy spectrum diagrams of ternesite at different times of wet carbonation.

matrix showed distinct cracking [69]. Moreover, the CC matrix tended to grow into a cluster structure, similar to findings by other scholars [34]. This phenomenon also provided nucleation sites for the subsequent large-scale formation of aragonite.

After 15 min of reaction, micrometer-scale aragonite whiskers and plate-like gypsum were clearly observed (Fig. 12(f)). Given that CC is less soluble than gypsum, the appearance of a large number of gypsum crystals indicates that the reaction was almost complete and the carbonate content in the system was relatively stable. Quantitative XRD analysis revealed that the system contained 30 wt% calcite, although only a small number of calcite particles were observed on the matrix surface. In addition, the cracking of the nucleation matrix and the transition of aragonite whiskers from aggregation to dispersion were clearly evident in Fig. 12(f). Notably, Fig. 12(g) shows particles with smooth surfaces, which, according to EDS analysis, contained high Si content, minor Ca, and negligible S. During carbonation, Ca and S from ternesite were largely leached and reprecipitated in solution, while Si—O bonds continuously polymerized inward to form a gel [45]. Therefore, these particles are considered to be decalcified and desulfurized ternesite, primarily composed of Si-gel. As the reaction time extended, aragonite crystals continuously crystallized and grew, with particle size increasing from 0.8 to 1 μm to 1–2 μm, and a mixture of aragonite and Si-gel was observed (Fig. 12(h)). The preferential formation of aragonite over calcite by ternesite may facilitate Si-gel leaching [70]. After 120 min, aragonite, calcite, gypsum, and Si-gel were clearly identifiable as carbonation products in Fig. 12(i).

3.3.2. Phase distribution

Fig. 13 illustrates the phase composition and spatial distribution of ternesite at various stages of the carbonation reaction. Variations in brightness in the BSE images correspond to different phases, which can be further distinguished by correlating with EDS elemental mapping [50]. In samples from the early reaction stage, the BSE image displayed a uniform bright white appearance, with Ca, Si, and S elements homogeneously distributed (Fig. 13(a)). As the carbonation progressed, the initially uniform bright white region evolved into a more complex phase distribution, with the white areas corresponding to CC precipitation and the gray regions likely representing polymerized Si-gel (Fig. 13(b)). During this process, significant diffusion and segregation of Ca and Si elements within ternesite were observed. CC predominantly formed in the outer layer of the ternesite particles, likely because carbonation initially occurred at the material's surface, accompanied by the outward migration of Ca. In contrast, Si aggregated in the inner region, resulting in the formation of a denser structure [34,66]. This distribution pattern indicates that the decomposition and reorganization of ternesite during carbonation are staged processes, with distinct migration and precipitation rates for different elements [45].

In Fig. 13(c), the boundary between CC and Si-gel become more distinct as the carbonation degree increased. The enlarged region highlights the distribution of Si-gel and CC, and the presence of aragonite is clearly delineated by the orange outline. The existence of needle-like aragonite whiskers may increase the porosity of the CC layer, which reduces its encapsulation effect on the internal Si-gel [70]. This structural change may facilitate the leaching of Si-gel, potentially improving the suitability of carbonated ternesite as a SCM [71]. EDS mapping further confirmed the distribution differences of Ca and Si, and also showed the distribution of O, C, and S. The S in the ternesite structure was released and subsequently reacted with Ca in the solution to form gypsum. This process not only affected the phase composition of ternesite, particularly influencing the growth of different CC, but also potentially impacted the microstructure and macroscopic properties of the material [45]. The disappearance of the S—O characteristic peak of ternesite and the appearance of the gypsum characteristic peak in the FT-IR analysis of Section 3.2.1 further corroborated this conclusion. After 120 min of reaction, the carbonation products became intermixed, and the phase boundaries appeared increasingly diffuse (Fig. 13(d)). In the later stages of the reaction, continued CO_2 introduction induced repeated dissolution–reprecipitation cycles of CC, promoting transformation toward more stable crystalline forms. Consequently, the CC shell may have been disrupted, facilitating the release of internal Si-gel. The presence of separate Si-gel was also detected in Fig. 13(d). In summary, the formation and separation of CC and Si-gel, the presence of aragonite whiskers, and the potential leaching of Si-gel, as observed in Fig. 13, may all significant impact the properties and applications of ternesite-based materials.

3.3.3. True density evolution

Fig. 14 illustrates the evolution of ternesite true density throughout the wet carbonation process. The green dots in the figure depict the trend of true density as a function of carbonation time. Additionally, the reference values of true density for ternesite, aragonite, and calcite are presented, along with the corresponding SEM images of the samples. During the initial 10 min, the true density decreased markedly from 2.99 g/cm^3 to 2.65 g/cm^3. This reduction is attributed to the breakdown of the ternesite structure and the resulting increase in porosity induced by the carbonation reaction. Additionally, the formation of low-density phases, such as CC and gypsum, as along with increased porosity resulting from Si-gel polymerization, further contributed to the decrease in true density [47]. The 10 min mark represents a critical transition point, where the primary phase shifted from ternesite to CC. Between 10 and 60 min, the true density increased, reaching 2.95 g/cm^3. This trend suggests that, as the carbonation reaction proceeded, CC crystals grew rapidly and filled the pores, thereby increasing the material's density. At 60 min, the true density approached that of aragonite, indicating that aragonite had become the predominant crystalline form of CC at this stage. After 60 min, the true density decreased again and eventually stabilized at 2.83 g/cm^3. This decrease may be attributed to the weakening of the precipitation and pore-filling effects of CC, accompanied by the transformation of aragonite to calcite and the leaching of Si-gel, all of which could have affected the true density [56,72]. Overall, variations in true density indirectly reflect changes in phase composition and volumetric properties of the system.

3.3.4. Pore structure

The effect of carbonation treatment on the pore structure of ternesite was studied using BET analysis. Figs. 15 and 16 show the N_2 adsorption-desorption isotherms and BJH adsorption-desorption pore volume change curves of six groups of ternesite with different carbonation

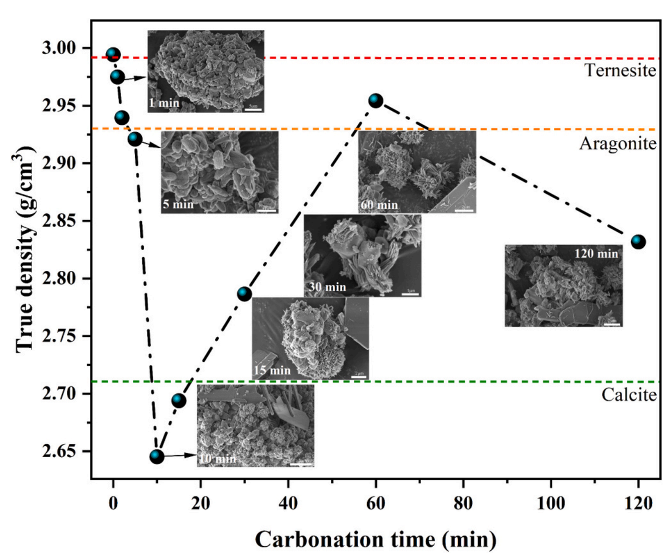

Fig. 14. True density and corresponding morphology of ternesite at different times of wet carbonation.

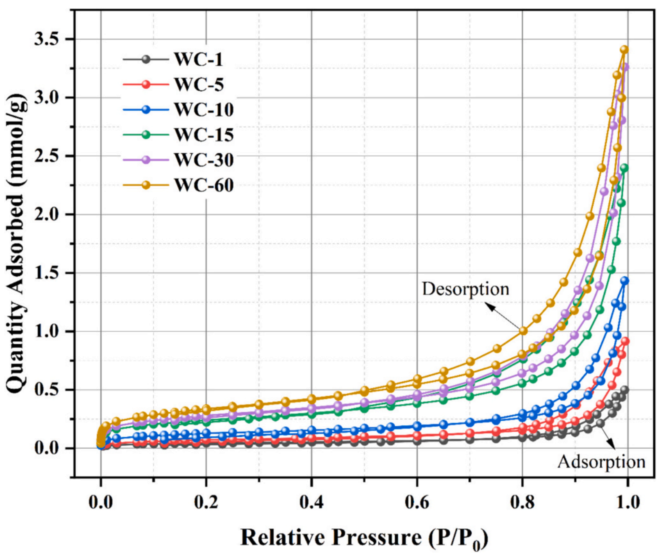

Fig. 15. N_2 adsorption-desorption isotherm of ternesite at different times of wet carbonation.

periods. The trend and shape of the N_2 adsorption–desorption isotherm curves in Fig. 15, according to the classification and nomenclature of the International Union of Pure and Applied Chemistry (IUPAC) [73], indicate that all the isotherms of the samples belong to type IV. This type is typically associated with mesoporous materials (For reference, pores with sizes <2 nm as micropores, 2–50 nm as mesopores, and > 50 nm as macropores). The observed isotherms indicate capillary condensation, especially at higher relative pressures (P/P_0 approaching 1), where a sharp increase in adsorption occurs. A pronounced hysteresis loop is evident between relative pressures of approximately 0.4 and 0.9, reflecting the difference between adsorption and desorption branches. This loop may be similar to an H_2-type hysteresis loop, further confirming the mesoporous nature of the material. As the carbonation time increased, the adsorption capacity of the samples gradually increased, indicating an enhancement in both specific surface area and pore volume. At 15 min, the isotherm was significantly higher than at 10 min, indicating that the pore structure of the sample had changed significantly at this moment. This transformation is primarily attributed to the rapid growth of CC, which altered the pore structure [74].

Fig. 16 presents the BJH pore volume distribution diagrams of the samples, including both adsorption and desorption pore volume distributions. Differences in pore diameter distribution of ternesite at various carbonation times are illustrated in Figs. 16(a) and (c). The most

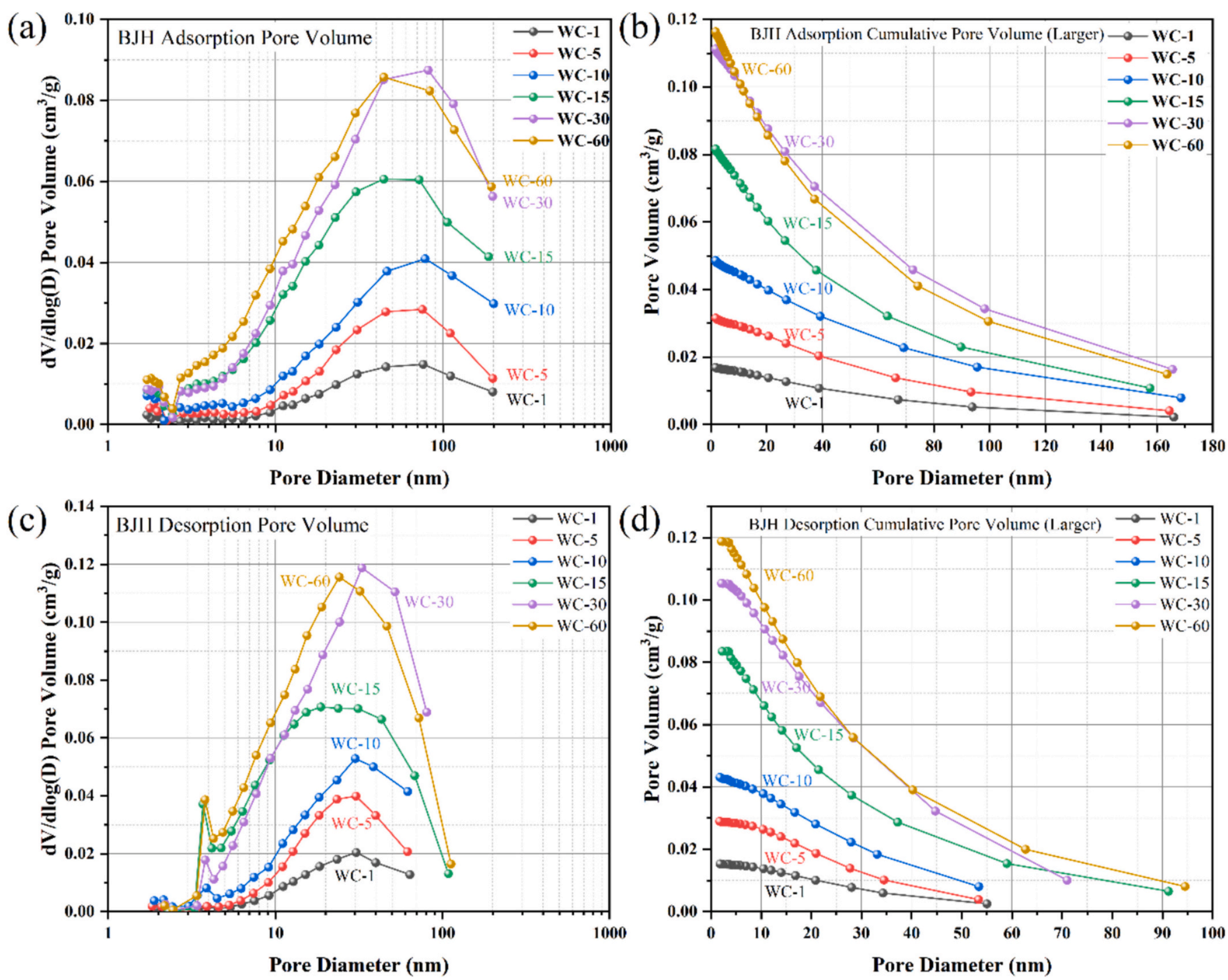

Fig. 16. BJH pore volume of ternesite at different times of wet carbonation; (a) BJH adsorption pore volume; (b) BJH adsorption cumulative pore volume; (c) BJH desorption pore volume; (d) BJH desorption accumulation pore volume.

probable pore diameter is defined as the pore diameter with the highest probability of occurrence in the pore diameter distribution. As shown in Fig. 16(a) and (c), all samples exhibited a distinct peak in the range of 20–40 nm, representing their most probable pore diameter. Fig. 16(b) and (d) illustrate the relationship between cumulative pore volume and pore diameter. The cumulative pore volume of the samples increased with the carbonation reaction time. Table 5 lists the detailed pore structure data of the carbonated samples the samples. Both the BET surface area and total pore volume of the carbonated samples increased by approximately eightfold with increasing carbonation degree from 1 min to 60 min. This increase was inextricably linked to the disruption of the ternesite structure and the generation of carbonation products. The most probable and average pore diameters of the samples both decreased with the reaction time, indicating that the carbonation treatment refined the pore diameter of the samples, mainly due to the formation of microcrystalline CC [75]. However, the most probable and average pore diameters increased significantly after 30 min. Previous analysis indicates that the main crystal type of CC transitions from calcite to aragonite at 30 min of reaction. Subsequently, aragonite continues to grow and crystallize, reaching a content as high as 45 wt% at 60 min. Therefore, the observed changes in the pore structure are attributed to the increase and growth of aragonite. In conjunction with solution chemistry and morphological analyses, these findings suggest that aragonite formation impedes the structural densification of carbonation products, while facilitating Si-gel leaching and CO_2 transport.

4. Discussion

In this study, the phase composition and microstructural evolution of ternesite at various stages of the wet carbonation reaction were systematically analyzed, and the carbonation reaction kinetics were characterized and simulated. By integrating relevant reports on ternesite and other silicate minerals [33,34,37,45], it was determined that the carbonation reaction of ternesite primarily involves the dissolution of CO_2 and ternesite, followed by the formation of CC, gypsum, and Si-gel. The related dissolution and precipitation reactions are presented in Eqs. (10) to (16). Notably, although ternesite has demonstrated good reactivity in the aluminate environment, its hydration activity remains to be fully elucidated (Eqs. (9) and (12)) [32]. Specifically, during the wet carbonation process, intermediate products such as CH and C-S-H were not detected. Instead, CC was formed directly via the reaction between Ca^{2+} and CO_3^{2-} (Eq. (14)), and Si—O units were directly polymerized to form Si-gel.

$$4CaO{\cdot}2SiO_2{\cdot}CaSO_4 + 8H_2O \rightarrow 5Ca^{2+} + 8OH^- + SO_4{}^{2-} + 2H_4SiO_4 \tag{10}$$

$$Ca^{2+} + SO_4{}^{2-} + 2H_2O \rightarrow CaSO_4{\cdot}2H_2O \tag{11}$$

$$2Ca^{2+} + 4OH^- + H_4SiO_4 \rightarrow C-S-H(gel) + Ca(OH)_2 \tag{12}$$

$$CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow HCO_3{}^- + H^+ \rightarrow CO_3{}^{2-} + 2H^+ \tag{13}$$

Table 5
Surface area and pore volume of ternesite at different times of wet carbonation.

Samples	BET Surface Area (m^2/g)	Maximum Quantity Adsorbed (mmol/g)	Total Pore Volume (cm^3/g)	Most Probable Pore Diameter (nm)	Average Pore Diameter (nm)
WC-1	3.51	0.50	0.02	30.52	18.67
WC-5	5.81	0.91	0.03	30.34	18.56
WC-10	10.31	1.43	0.04	30.08	17.19
WC-15	19.23	2.40	0.08	18.71	15.62
WC-30	22.33	3.26	0.11	32.92	19.05
WC-60	27.07	3.41	0.19	24.17	16.91

$$Ca^{2+} + CO_3{}^{2-} \rightarrow CaCO_3\downarrow \tag{14}$$

$$H_4SiO_4 + (n-2)H_2O \rightarrow SiO_2{\cdot}nH_2O(gel) \tag{15}$$

$$4CaO{\cdot}2SiO_2{\cdot}CaSO_4 + 4CO_2 + (2+n)H_2O \rightarrow 4CaCO_3 + CaSO_4{\cdot}2H_2O + 2SiO_2{\cdot}nH_2O \tag{16}$$

It is well established that polycrystalline CC, including calcite, aragonite, and vaterite, is an inevitable product of the carbonation of Ca-containing minerals [76]. Calcite is the most stable and predominant form of CC, while aragonite and vaterite are structurally unstable and generally rarely produced or converted to calcite as the carbonation reaction proceeds. Consequently, formation and persistence of metastable CC phases depend on specific carbonation conditions, including temperature, pH, humidity, and CO_2 pressure [77]. Additionally, crystalline modifiers such as Mg^{2+} and amino acids are commonly employed to direct the carbonation reaction toward the formation of aragonite or vaterite [41,42]. However, in this study, the nucleation and growth of CC during the wet carbonation of ternesite displayed distinct stage-dependent characteristics, with the crystal morphology predominantly favoring aragonite formation.

4.1. *Carbonation behavior of ternesite from the perspective of the crystal transformation process of calcium carbonate*

Table 6 lists the CC content in the samples, as determined by XRD and TG results. A discrepancy exists between the CC contents obtained by the two methods (△CC), with TG yielding higher values than XRD. This discrepancy arises because XRD analysis only quantifies CC crystals, whereas TG results also include amorphous and poorly crystallized CC [10]. Consequently, the △CC values can reflect the trends of ACC and PCCC in the system.

Fig. 17 depicts the evolution of CC crystallinity and crystallite size throughout the carbonation reaction. Initially, calcite was detected with a crystallinity of 89.7% and a crystallite size of 36.0 nm. After 15 min, these values increased to 94.9% and 48.7 nm, respectively. At 30 min, the crystallinity of calcite dropped sharply to 88.2%, while its crystallite size rapidly increased to 73.8 nm. At 120 min, the crystallinity of calcite increased slightly, while its crystallite size decreased slightly. Aragonite first appeared at 10 min, with its crystallinity and crystallite size continuously increased from 59.5% and 45.6 nm to 88.2% and 67.1 nm, respectively. Notably, the crystallite size of aragonite decreased at 60 min. Based on the results in Section 3, combined with the contents of Table 6 and Fig. 17, the crystal growth process of the main carbonation product CC can be summarized as follows:

Stage 1 (0–10 min): During the initial reaction phase, the dissolution of CO_2 and ternesite synergistically increased the concentrations of Ca^{2+} and CO_3^{2-} in the solution (Fig. 11). This provided a sufficient basis for CC nucleation. During this period, the solution pH remained relatively high, with approximately 80% of the carbonation reaction occurring in an alkaline environment ($pH > 8$). This condition favored the formation of ACC and metastable CC [58]. The high supersaturation led to a rapid nucleation rate of CC, resulting in small crystallite sizes [77]. Therefore, the early-formed CC exhibited high reactivity and small crystallite size, serving as effective nucleation sites for subsequent CC precipitation. Additionally, the morphology of CC changed, consistent with findings from other studies [47]. Initially, nanoparticles with an ACC-like morphology covered the nucleation matrix. As the crystals grew and underwent polycrystalline transformations, the matrix contours became more distinct, accompanied by cracking (Fig. 12(a)-(d)).

Stage 2 (10–30 min): △CC reached a maximum at 10 min then decreased, indicating the transition of ACC and poorly crystallized CC to thermodynamically more stable crystalline forms. At this stage, the total CC crystal content stabilized, while crystallinity and crystallite size increased with carbonation time. This period was characterized by CC polycrystalline growth. Calcite was the predominant CC phase, although

Table 6
$CaCO_3$ content of carbonated ternesite by XRD Rietveld method and TG analysis.

Items	Content $CaCO_3$ (wt%)							
	WC-1	WC-2	WC-5	WC-10	WC-15	WC-30	WC-60	WC-120
Results of XRD Rietveld method	5.67	7.67	11.22	35.61	45.43	50.33	51.07	51.78
Results of TG analysis	10.62	10.81	16.10	48.09	57.55	57.92	57.65	58.03
Difference (△CC)	4.96	3.14	4.88	12.47	12.12	7.58	6.58	6.25

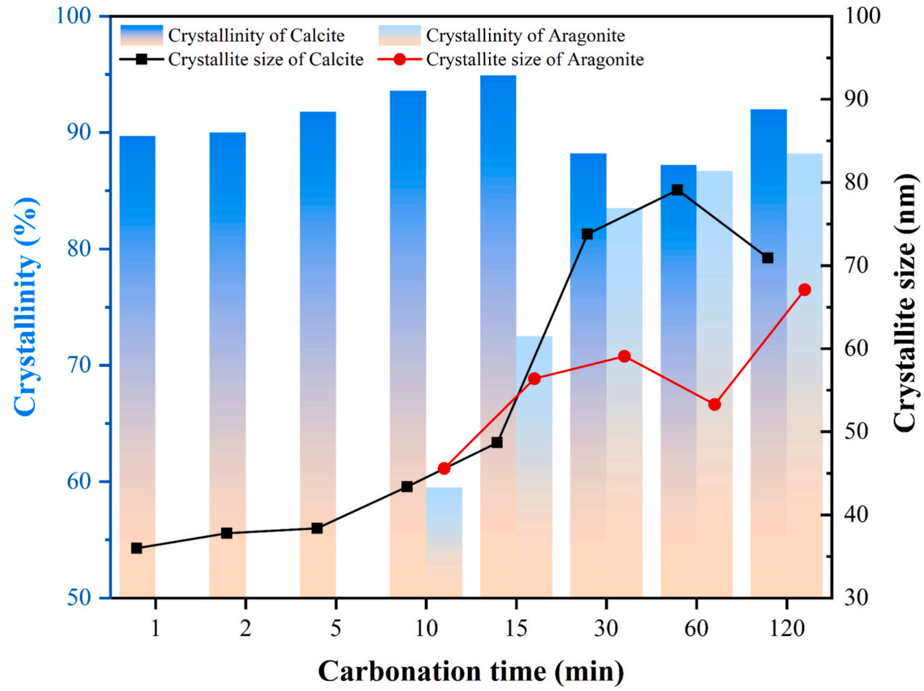

Fig. 17. Crystallinity and crystallite size of calcite and aragonite.

distinct calcite particles were not observed in SEM images. Instead, the nucleation matrix exhibited a clustered structure, with growth morphology favoring aragonite. As crystals grew and the matrix cracked, aragonite whiskers became clearly visible (Fig. 12(e)-(g)).

Stage 3 (30–120 min): When the reaction reached 30 min, the calcite content decreased while the aragonite content increased rapidly. During this period, CC underwent crystalline transformation during the dissolution-re-precipitation process, as evidenced by the decrease in the crystallinity of calcite in Fig. 17. The reduction in aragonite crystallite size at 60 min may be attributed to the incomplete crystallization of newly formed aragonite. In addition, the unit cell parameters and major interplanar spacings of calcite and aragonite are listed in Table 7. The (104) interplanar spacing of calcite increased as the reaction proceeded and decreased after 30 min, while aragonite showed a decrease in the d-spacing of all its (111) crystal planes after 10 min. These observations suggest that the main diffraction peak for calcite shifted to a lower angle, while that for aragonite shifted to a higher angle. The interplanar spacing of the crystal faces of the two types of CC changed, which also demonstrated the possibility of inter transformation during the growth of polycrystalline CC. However, as the reaction time further increased, the content and crystallinity of calcite in the sample at 120 min increased. Calcite is thermodynamically more stable than aragonite and is more readily formed under ambient conditions [77]. Therefore, in the later stages of the reaction, continued CO_2 supply facilitated the transformation of aragonite into calcite.

Throughout the experiment, CC evolved from amorphous and poorly crystalline forms in the early stages to aragonite and calcite in later stages. This evolution was closely related to the crystal structure of

Table 7
d-spacings and unit cell parameters of calcite and aragonite.

Sample	d-spacings (Å)		Unit cell parameters (Å)					Cell volume (Å^3)		Density (g/cm^3)	
			Calcite		Aragonite			Calcite	Aragonite	Calcite	Aragonite
	104	111	a	c	a	b	c				
WC-1	3.03	–	4.97	17.09	–	–	–	365.22	–	2.73	–
WC-2	3.03	–	4.97	17.10	–	–	–	366.37	–	2.72	–
WC-5	3.04	–	4.97	17.12	–	–	–	368.81	–	2.70	–
WC-10	3.04	3.41	4.98	17.18	5.01	7.67	5.70	369.48	218.90	2.70	3.04
WC-15	3.05	3.41	4.98	17.20	4.97	7.99	5.76	369.27	229.21	2.70	2.90
WC-30	3.05	3.39	4.98	17.20	4.95	7.96	5.74	370.22	226.24	2.69	2.94
WC-60	3.04	3.39	4.98	17.15	4.87	8.18	5.78	368.71	230.20	2.71	2.89
WC-120	3.04	3.38	4.98	17.17	4.94	7.93	5.72	368.17	224.37	2.71	2.96

ternesite, solution chemistry, and the thermodynamic stability of CC.

4.2. Calcium carbonate crystal transformation accelerated silica gel leaching

During the carbonation process, the microstructure of ternesite changed considerably, especially the morphological transformation of CC, which had a great impact on the leaching of Si-gel, another important product. In the initial stages of the reaction, surface defects and pores generated by ternesite dissolution provided favorable sites for CC nucleation (Fig. 12) [56]. In addition, the Si—O bonds in the structure readily polymerized following the disruption of the ternesite crystal. After 10 min of carbonation, over 80% of the Si had transformed into Si-gel, and this proportion exceeded 95% after 15 min. As CC continued to deposit, a “core-shell” structure was formed, with the CC shell encapsulating the polymerized Si-gel and unreacted ternesite particles (Fig. 13). Several previous studies have reported this structure to some extent limited ion exchange, hindering the further progress of the carbonation reaction and the leaching of active Si-gel [35,78,79]. Various chemical and mechanical methods have been explored to obtain highly pozzolanic Si-gel or value-added nano-SiO_2 during mineral carbonation [44,58]. However, these approaches raised production costs and increased operational complexities. In this study, it was observed that different crystalline types of CC significantly affected the pore structure and the leaching of Si-gel. As shown in Fig. 11 and discussed in Section 3.3, the formation of aragonite increased porosity, which facilitated the leaching of Si-gel and the transport of CO_2. On the one hand, the dissolution-reprecipitation of CC to form aragonite disrupted the integrity of the original CC shell. On the other hand, the needle-like structure of aragonite led to an increase in the porosity of the CC layer in the “core-shell” structure, thereby reducing its encapsulation effect on the inner Si-gel and making it easier for Si-gel to leach out of the material. The crystalline transformation of CC has broken the “core-shell” structure and facilitated the leaching of Si-gel, which was one of the important innovations in this study. This phenomenon may enhance the potential of carbonated ternesite as a SCM and offers new strategies for overcoming the carbonation barrier. Nevertheless, the mechanisms underlying the crystalline transformation of CC during wet carbonation of ternesite, particularly the predominance of metastable aragonite, require further investigation.

4.3. Mechanism of wet carbonation in ternesite

Based on the above analysis, the carbonation mechanism of ternesite is discussed, and a schematic diagram is presented as shown in Fig. 18. Ternesite crystals belong to the orthorhombic crystal system, the Pnma space group, which is characterized by $[SiO_4]$ tetrahedra and $[SO_4]$ tetrahedra connected via Ca cations to form a three-dimensional network exhibiting a distinctly layered structural arrangement [23]. In terms of elemental composition, ternesite contains an additional SO_4^{2-} anion compared to silicate minerals, which plays a critical role in maintaining structural charge balance. This compositional feature leads to distinct types and concentrations of anions released during carbonation, differentiating ternesite from conventional silicate minerals. The presence of SO_4^{2-} may alter the chemical properties of the solution, thereby influencing the crystalline morphology of CC [80–82].

Previous studies [83] have indicated that the crystal structure of minerals, especially the Ca coordination number and Ca—O bond length, governs their reactivity with CO_2 and the nature of the resulting reaction products. Table 8 lists the atomic coordination numbers and bond lengths for ternesite, gypsum, calcite, and aragonite. In ternesite, the Ca coordination number ranges from 6 to 7, with bond lengths varying between 2.32 Å and 2.70 Å. These relatively lower coordination numbers and shorter bond lengths facilitate the rapid release of Ca^{2+} into solution during the early stages of carbonation, thereby providing a sufficient supply of calcium for the formation of CC [84]. Simultaneously, the SO_4^{2-} ions coordinated with Ca^{2+} are likewise released into the solution. In contrast, the $[SiO_4]$ tetrahedron is highly stable, and the Si—O bonds tend to polymerize inwardly, forming layered or three-dimensional structures during the carbonation [60]. As a result, the Ca concentration in solution increases rapidly, whereas the Si concentration remains comparatively low. Consequently, the high saturation and high pH of the solution favor the rapid nucleation and growth of CC, particularly in the form of calcite [66]. Furthermore, Busenberg and Plummer's earlier report [85] indicated that the presence of SO_4^{2-} in solution competes with CO_3^{2-} and adsorbs onto active sites at the calcite crystal interface, thereby retarding calcite crystal growth. This observation aligns with the findings of the present study, in which ACC and calcite exhibiting inhibited crystal growth were the predominant products formed during the early stage of ternesite carbonation (0–10 min). The high supersaturation and rapid reaction kinetics place the system under kinetic rather than thermodynamic control. Product formation is governed by the kinetics of the reaction pathway (rapid generation of ACC and or nano-crystals) rather than ultimate thermodynamic stability (large calcite particles).

As ternesite underwent near-complete dissolution, the Ca^{2+} concentration and pH of the solution declined markedly (Fig. 11). Notably, the pH of the solution dropped below 9 within 10 min, inhibiting HCO_3^-

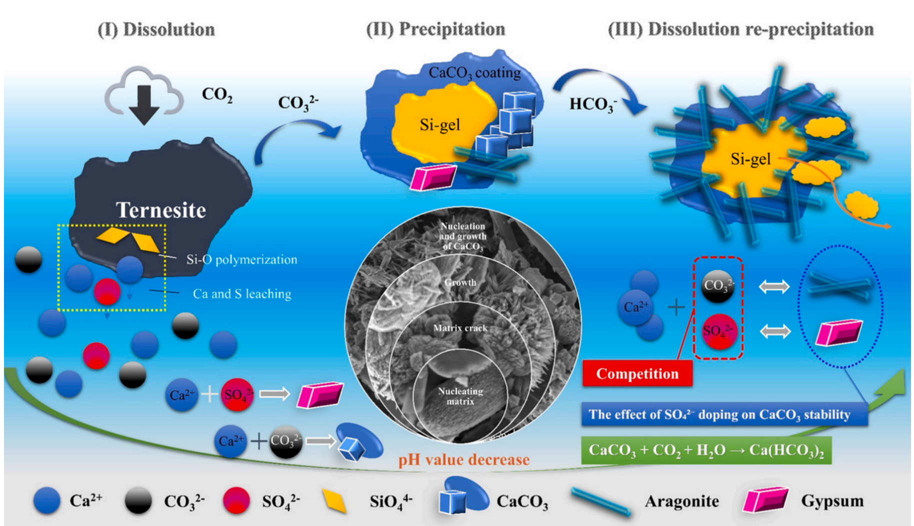

Fig. 18. Schematic mechanism of the carbonation reaction of ternesite.

Table 8
Coordination number and bond length of ternesite, calcite and aragonite.

Mineral	Ca-O		Si-O		S-O	
	Coordination number	Bond length (Å)	Coordination number	Bond length (Å)	Coordination number	Bond length (Å)
Ternesite[a]	6/7	2.32–2.70	4	1.65–1.66	4	1.49–1.50
Gypsum[a]	8	2.37–2.62	–	–	4	1.49–1.50
Calcite[b]	6	3.40	–	–	–	–
Aragonite[b]	9	2.53	–	–	–	–

[a] Reference to ICSD database;
[b] Reference [35].

dissociation. Consequently, the proportion of CO_3^{2-} declined, and HCO_3^- emerged as the dominant carbonate species. At this point, gypsum was produced in large quantities, suggesting that the concentration of CO_3^{2-} in the solution was significantly lower than that of SO_4^{2-}. The total amount of free water in solution decreased as the reaction proceeded, due to both the evaporation of water from the heat of reaction and the increase in chemically bound water in the gypsum. This reduction in available water further hindered CO_2 dissolution and dissociation. The amounts of CC and gypsum crystals continued to increase and eventually stabilized as carbonation progressed; however, the thermodynamically stable calcite partially transformed into aragonite.

On the one hand, as HCO_3^- increased and became dominant in the solution, the crystallized CC may undergo the reaction shown in Eq. (17) [58], and subsequently re-dissolve. On the other hand, SO_4^{2-} can replace CO_3^{2-} within the CC polycrystal structure. This substitution is thermodynamically favorable in vaterite, less favorable in calcite, and even more restricted in aragonite [86]. In other words, calcite crystals retain structural stability more readily than aragonite upon SO_4^{2-} incorporation. However, this reduced the solubility of calcite. Previous research indicates that calcite containing 3 mol% SO_4^{2-} exhibits a solubility product comparable to that of aragonite [85]. Consequently, CC experienced repeated dissolution-reprecipitation cycles, accompanied by transitions among different crystalline forms. Furthermore, the Ca coordination number and bond length in gypsum are closer to those in aragonite [84]. This chemical environment therefore favors the nucleation, growth, and stabilization of aragonite. In conclusion, the changes in ion ratios and especially the presence of SO_4^{2-} modified the thermodynamic properties of CC crystals. At this stage, the reaction was under combined thermodynamic and kinetic control, favoring the formation and stabilization of metastable CC, notably aragonite. After 120 min of carbonation, the aragonite content declined, while calcite content increased. During the later stage, carbonation was predominantly thermodynamically controlled, with CC transforming into its most stable polymorph, calcite.

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2 \tag{17}$$

5. Limitations and future perspectives

Overall, the dissolution of ternesite and the formation of polycrystalline CC during the reaction are closely related to the changes in the solution chemical environment. This phenomenon is fundamentally governed by the crystal structure, bonding characteristics, elemental composition, and unique carbonation products of ternesite. The present work has some limitations although it comprehensively investigated the carbonation behavior, kinetics and mechanism of ternesite. There are many other factors affecting the carbonation reaction, such as pH, temperature, and CO_2 concentration [87,88]. Further validation is required to ascertain the effect of reaction conditions on the ternesite carbonation reaction. This is necessary for subsequent targeted modulation of the carbonation product generation, especially the directed generation of different crystalline forms of CC. Additionally, while the effects of ternesite's crystal structure and bonding characteristics on the reaction were discussed, further data are required for comprehensive understanding. In addition, in light of the urgent demand for low-carbon building materials and the limitations of current SCM [14,71], ternesite will be further investigated as an SCM using wet carbonation technology. As demonstrated in this study, wet carbonation of ternesite yields substantial amounts of metastable CC, Si-gel, and gypsum, all of which exhibit high reactivity and are therefore suitable as SCM. Moreover, the intrinsic properties of ternesite can modulate aragonite formation and influence the leaching behavior of Si-gel. These findings may facilitate broader applications and increased interest in ternesite.

6. Conclusions

In this study, the carbonation behavior of the novel low-calcium mineral ternesite in aqueous solutions was systematically investigated, with a comprehensive investigation of the carbonation kinetic and mechanism at different stages. The experimental findings support the following conclusions:

1. Ternesite exhibited high carbonation reactivity and CO_2 sequestration capacity, achieving approximately 95% reaction completion within 15 min. After 120 min, the CO_2 sequestration capacity reached 342.8 g/kg, with a maximum carbonation degree of 96.1%. The final reaction products were aragonite, calcite, gypsum, and Si-gel. The carbonation kinetics of ternesite conformed to the surface-coverage model, leading to the development of a "core-shell" structure.
2. The aqueous carbonation of ternesite exhibited distinct kinetic stages: acceleration, deceleration, and stabilization. Mineral dissolution and CC nucleation occurred within the first 10 min, followed by continued precipitation and growth of carbonation products. The reaction stabilized after 30 min, coinciding with significant polycrystalline transformation of CC.
3. The mechanisms of CC nucleation, growth, and polycrystalline evolution during ternesite carbonation were clarified. Initially, amorphous and poorly crystalline CC formed, which subsequently transformed into calcite and aragonite. The final CC product exhibited a polycrystalline structure dominated by aragonite, characterized by low crystallinity and small crystallite size.
4. The CC produced underwent significant morphological changes, particularly with increased aragonite content, which disrupted the shell and enhanced the leaching of Si-gel. This was supported by observed variations in Si concentration and microstructural evolution.
5. Compared to silicate minerals, ternesite contains additional $[SO_4]$ groups for charge balance, resulting in distinct ionic release profiles during carbonation. This influences the crystallization morphology and stability of CC, with SO_4^{2-} promoting the formation and stabilization of metastable CC, particularly aragonite.

CRediT authorship contribution statement

Jixin Zhang: Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Kai Cui:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Jun Chang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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